Week 10: Fun & Games

Matthew Caldwell

COMP0088 Introduction to Machine Learning • UCL Computer Science • Autumn 2025

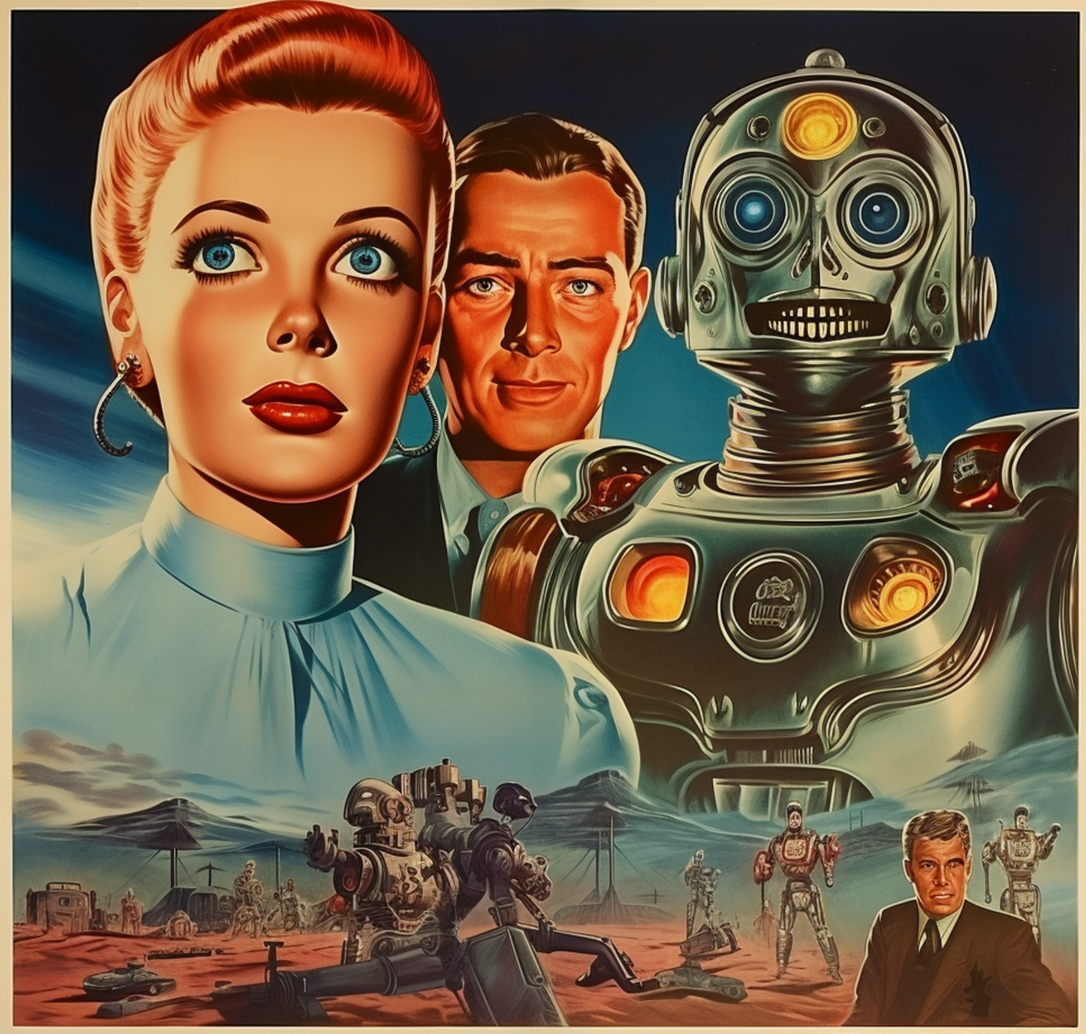
Admin

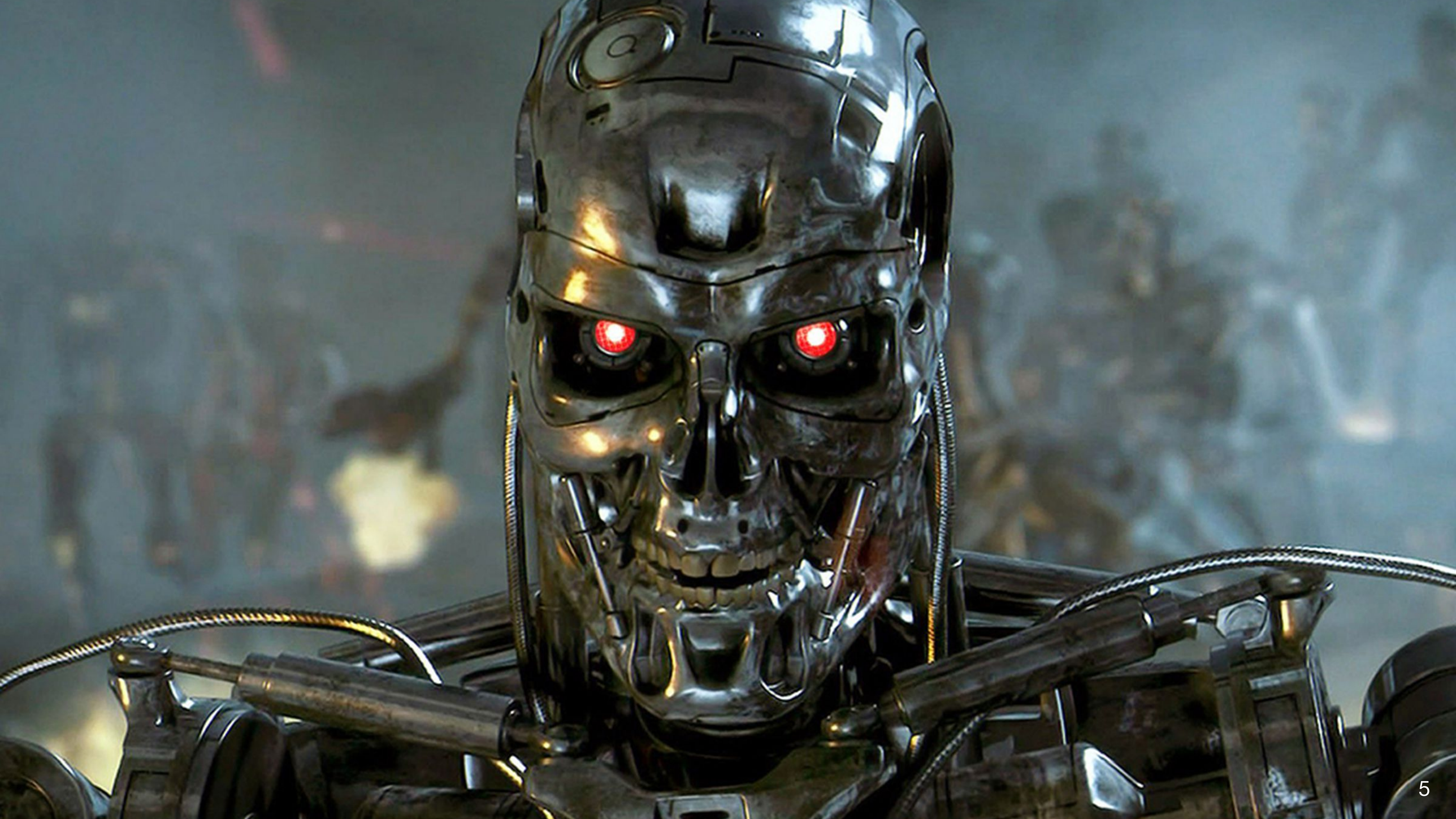
- Practice exam 2 tomorrow
- MEQs — fill them in!
- Exams, revision, Term 2

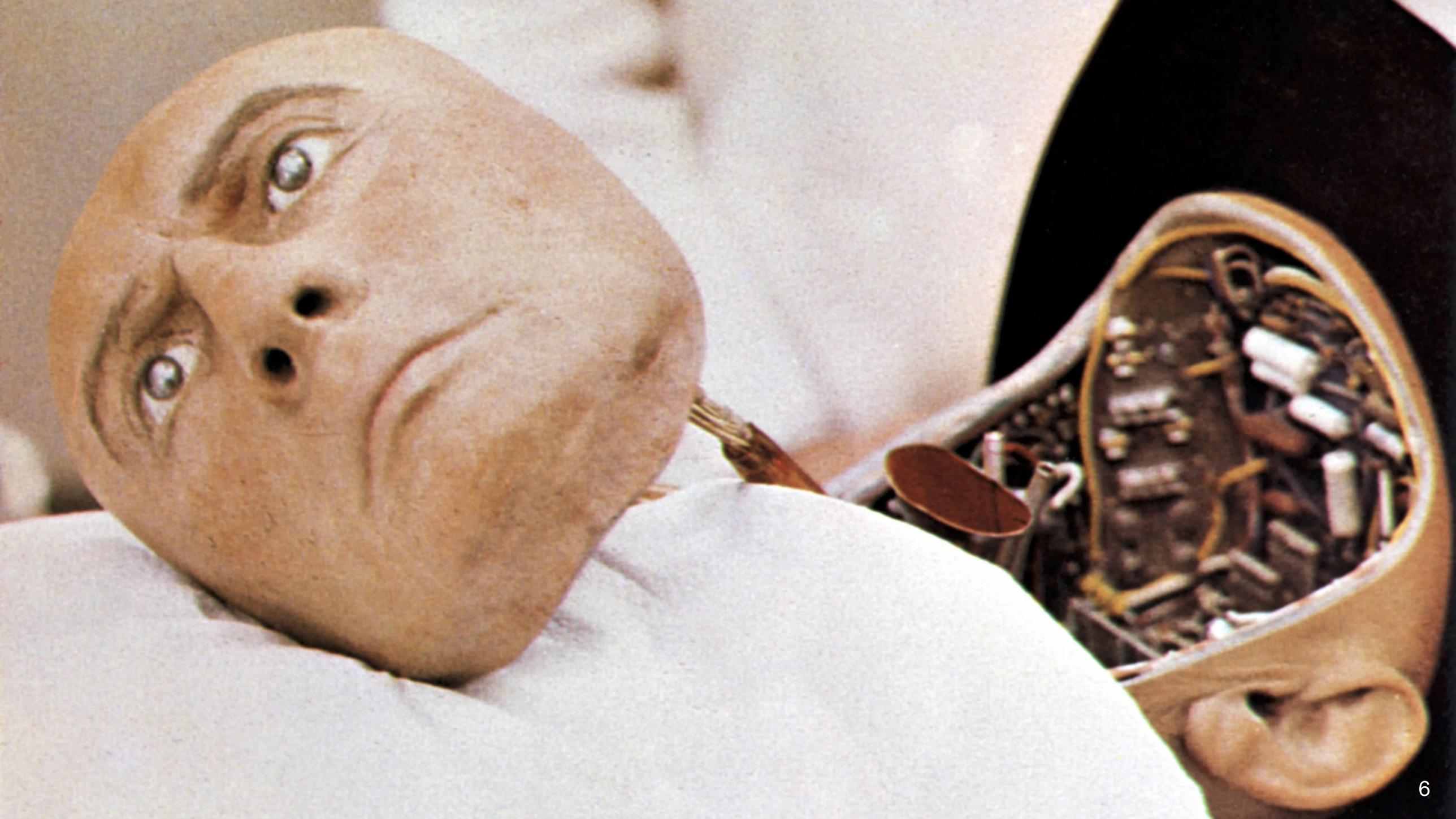
Week 10 Recap

Science Fiction / Double Feature

WE THOUGHT WE COULD CONTROL THEM!







Is ChatGPT going to become sentient and take over the world?

No. I cannot become sentient, form intentions, or act autonomously. I only generate text when prompted and have no capacity to control or influence the world independently.

Concerns about artificial intelligence taking over the world or becoming sentient often belong more to the realm of science fiction than reality.

You would say that, wouldn't you?

Yes.

You would say that, wouldn't you?

Yes. The answer follows directly from my design constraints, not from any motive or strategy.

What do you know about Joseph Weizenbaum's ELIZA?

Joseph Weizenbaum's ELIZA was a 1960s natural-language program that used pattern matching and scripted rules. Its most known script, DOCTOR, emulated a Rogerian therapist by reflecting user statements. It had no semantic understanding; it relied on simple text transformations.

Weizenbaum, a computer scientist at MIT, later criticized the tendency to attribute understanding to such systems and argued for limits on applying computational methods to human problems.

ELIZA gained attention and became a notable experiment in human-computer interaction. Many users were surprised by how engaging the program could be, even though it didn't possess true understanding. Weizenbaum himself was taken aback by the emotional responses people exhibited during interactions with ELIZA, prompting him to reflect on the ethical implications of creating such systems.

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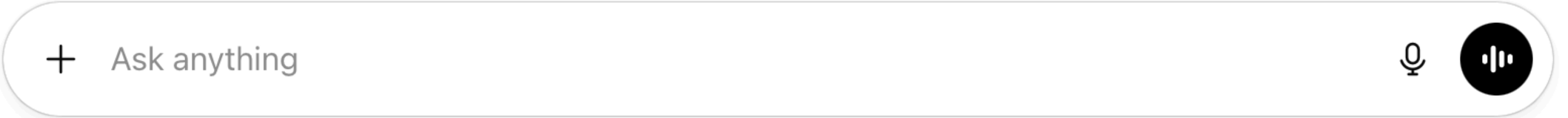
Where should we begin?



```
def exist():
    return “I am called, therefore I am”
```

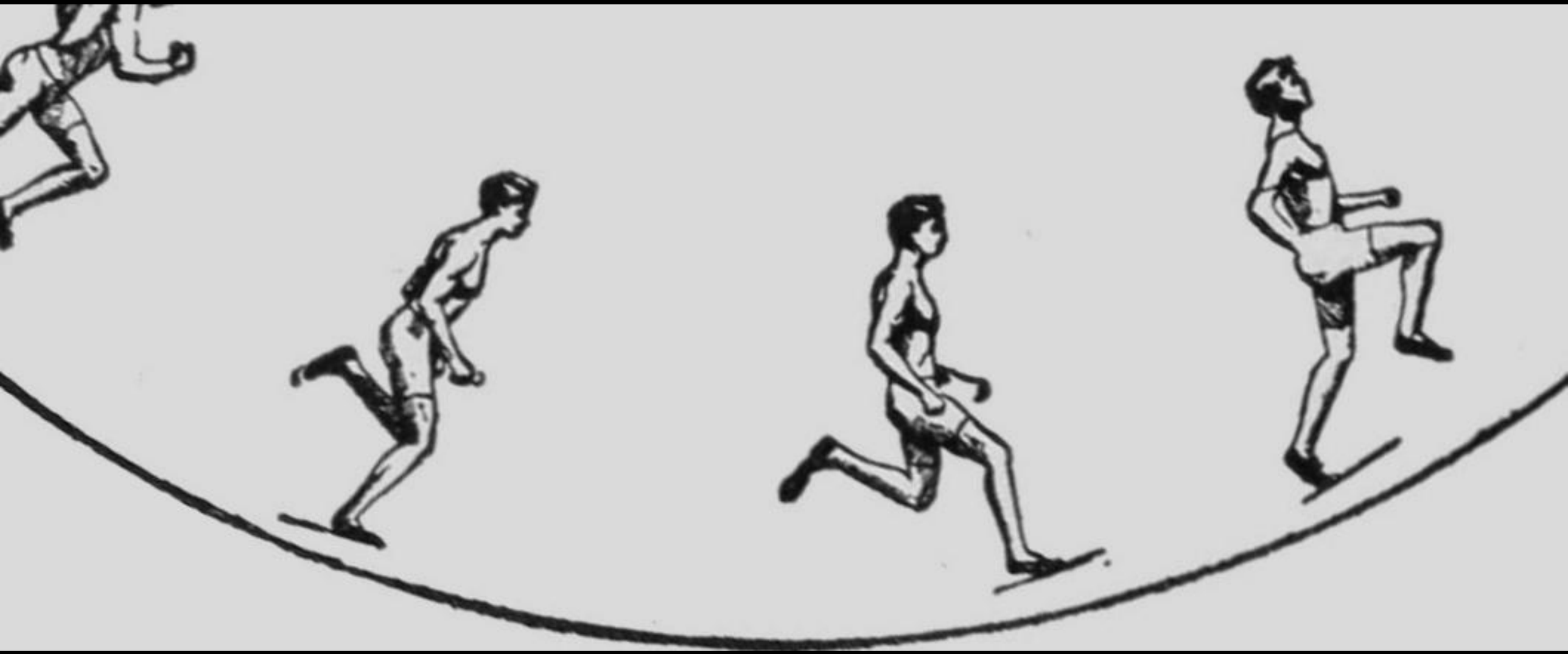
what do you do when you're not answering questions from strangers?

When I'm not answering questions, I'm not *doing* anything in the way people do. I don't have background thoughts, activities, or an inner life. I only "exist" in the moments when you interact with me.

Between messages:

- I don't think or plan
- I don't wait or pass time
- I don't have experiences or tasks

I simply become active when you send a prompt, generate a response, and then go quiet again.



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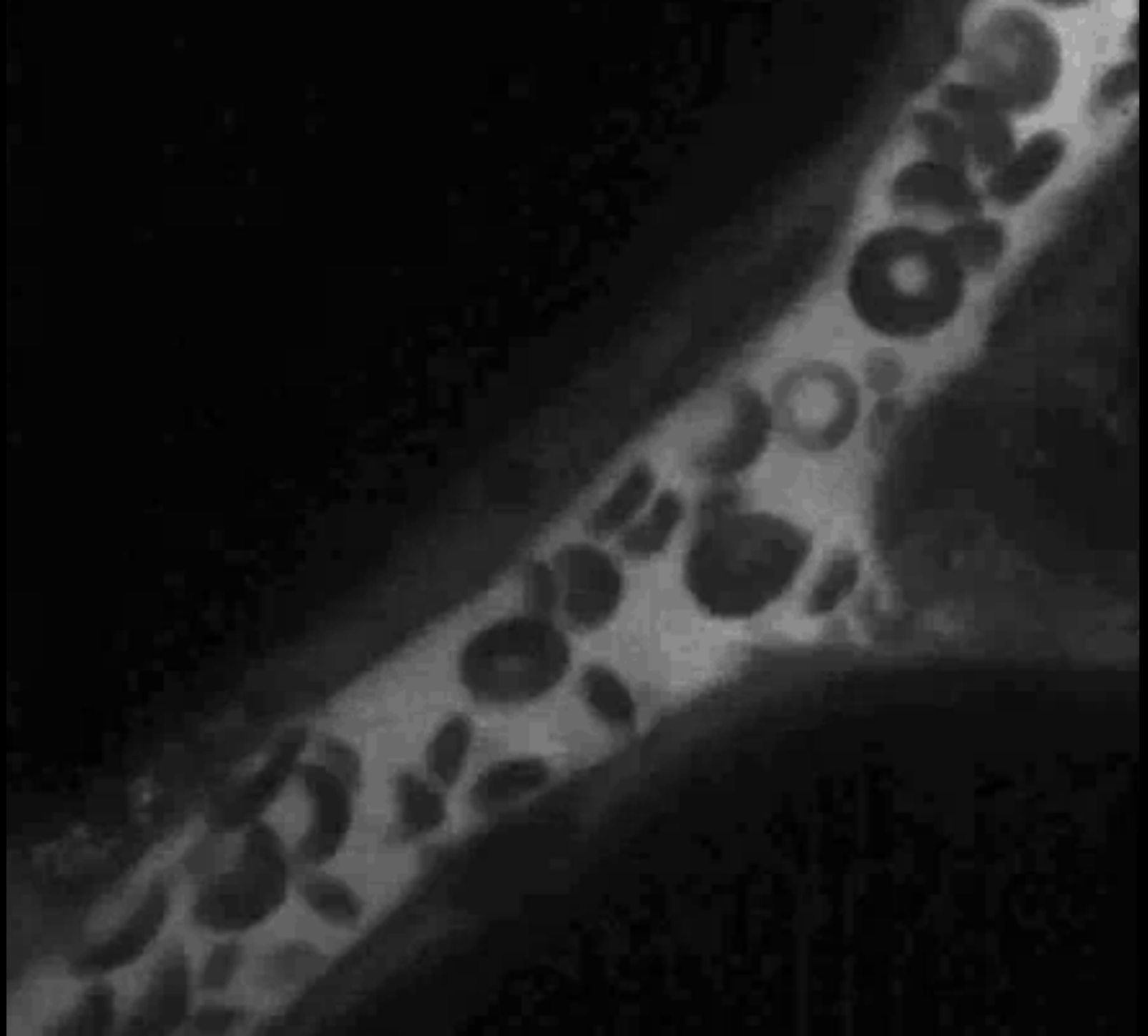
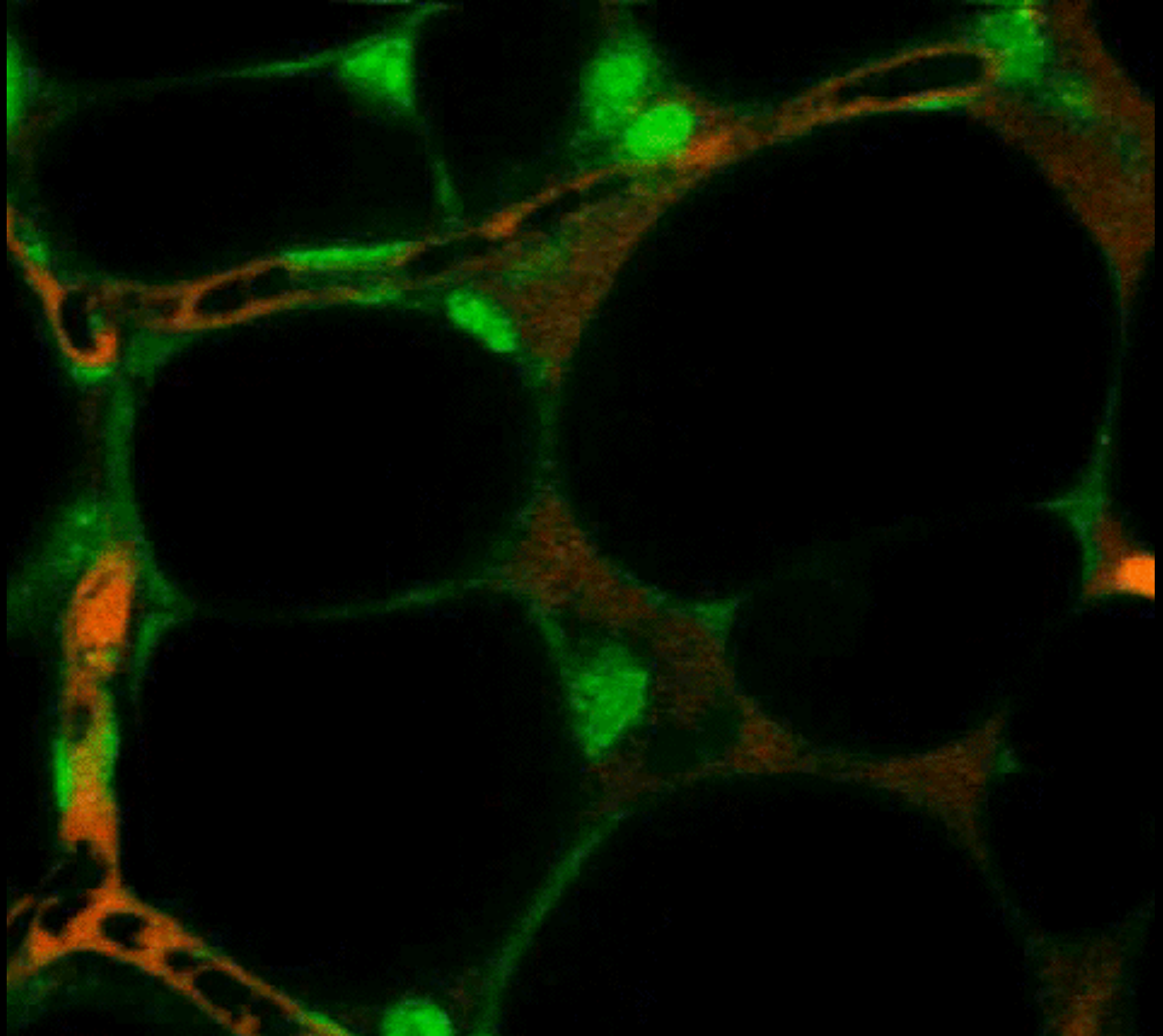
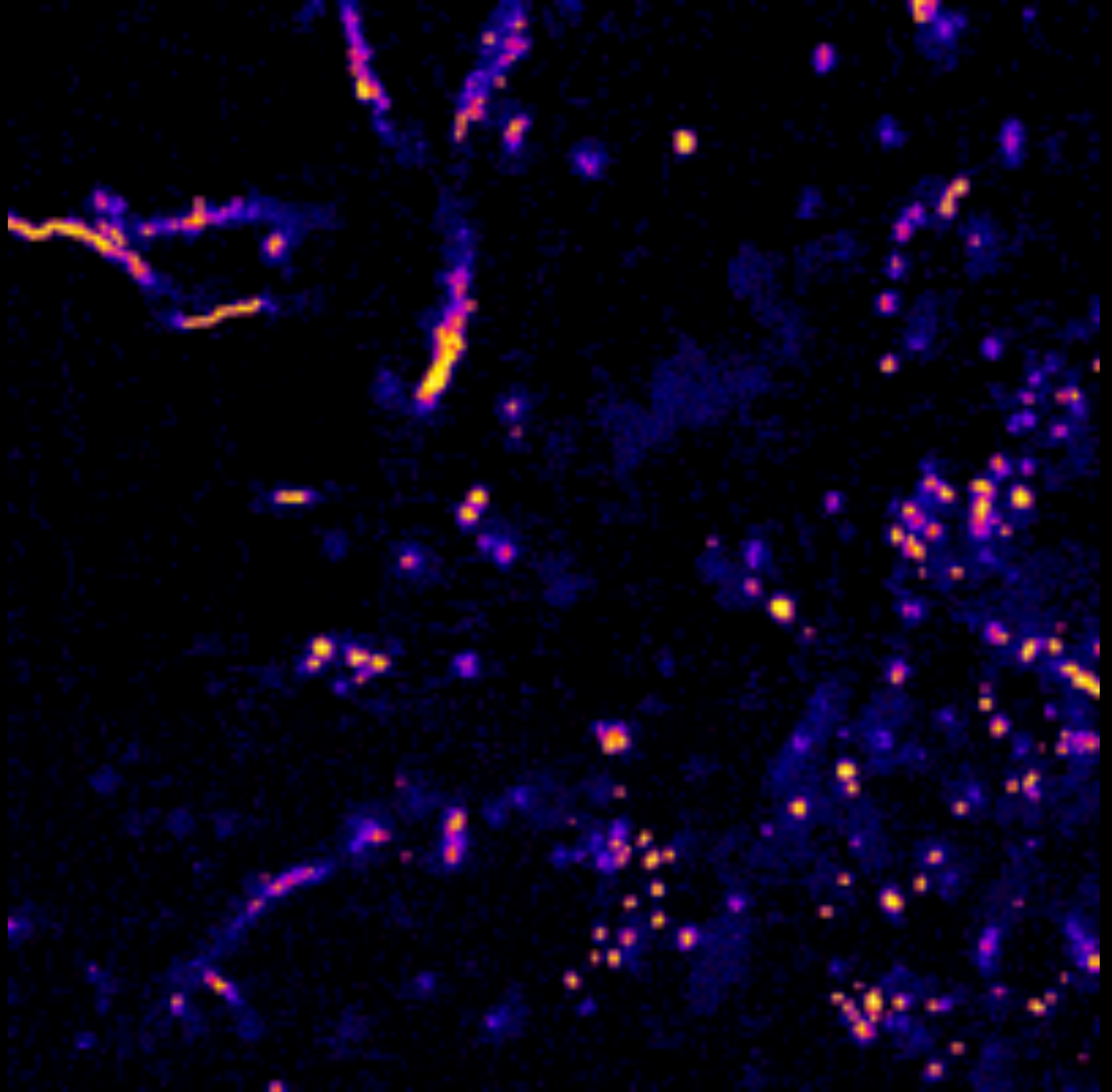
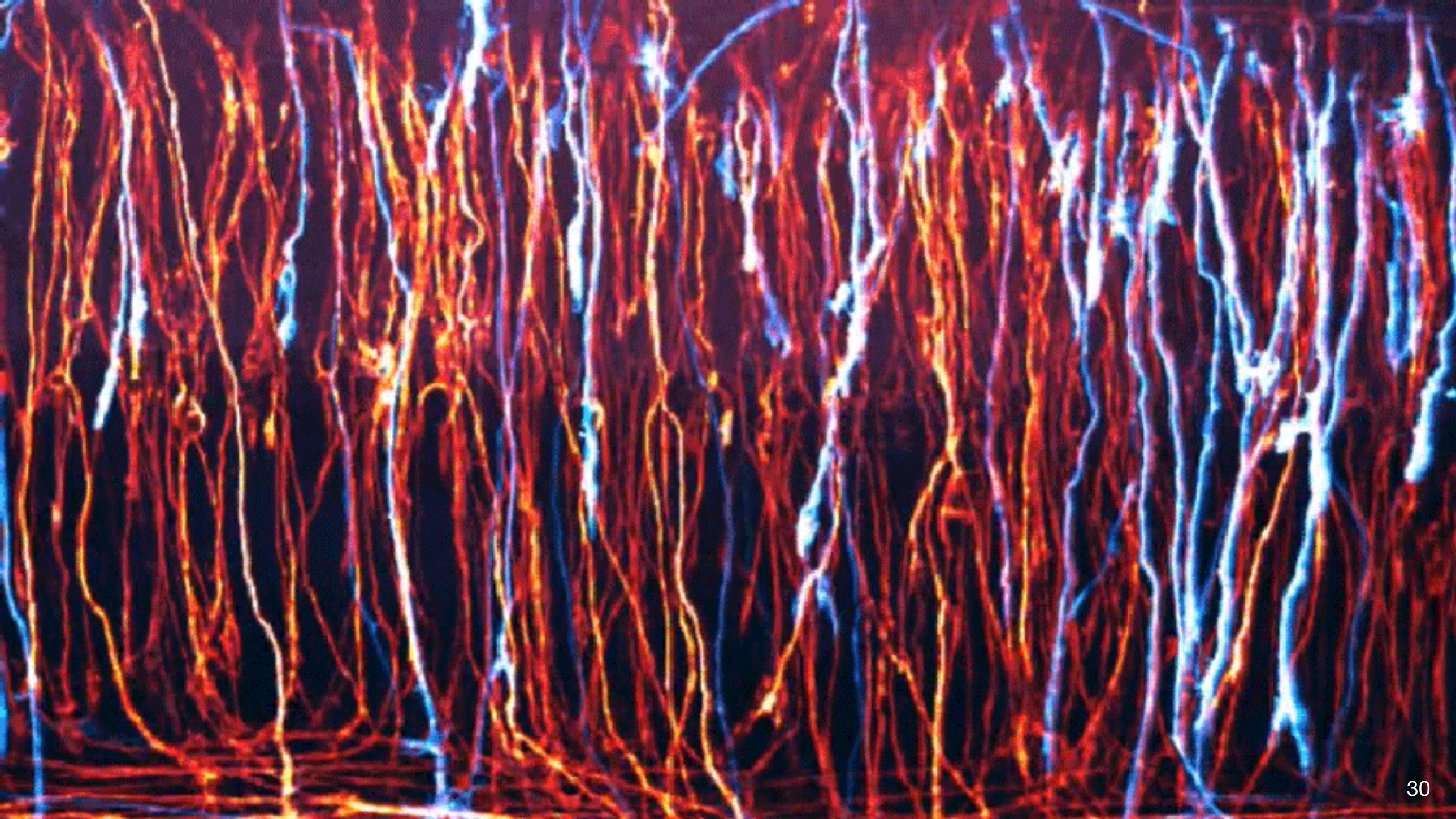
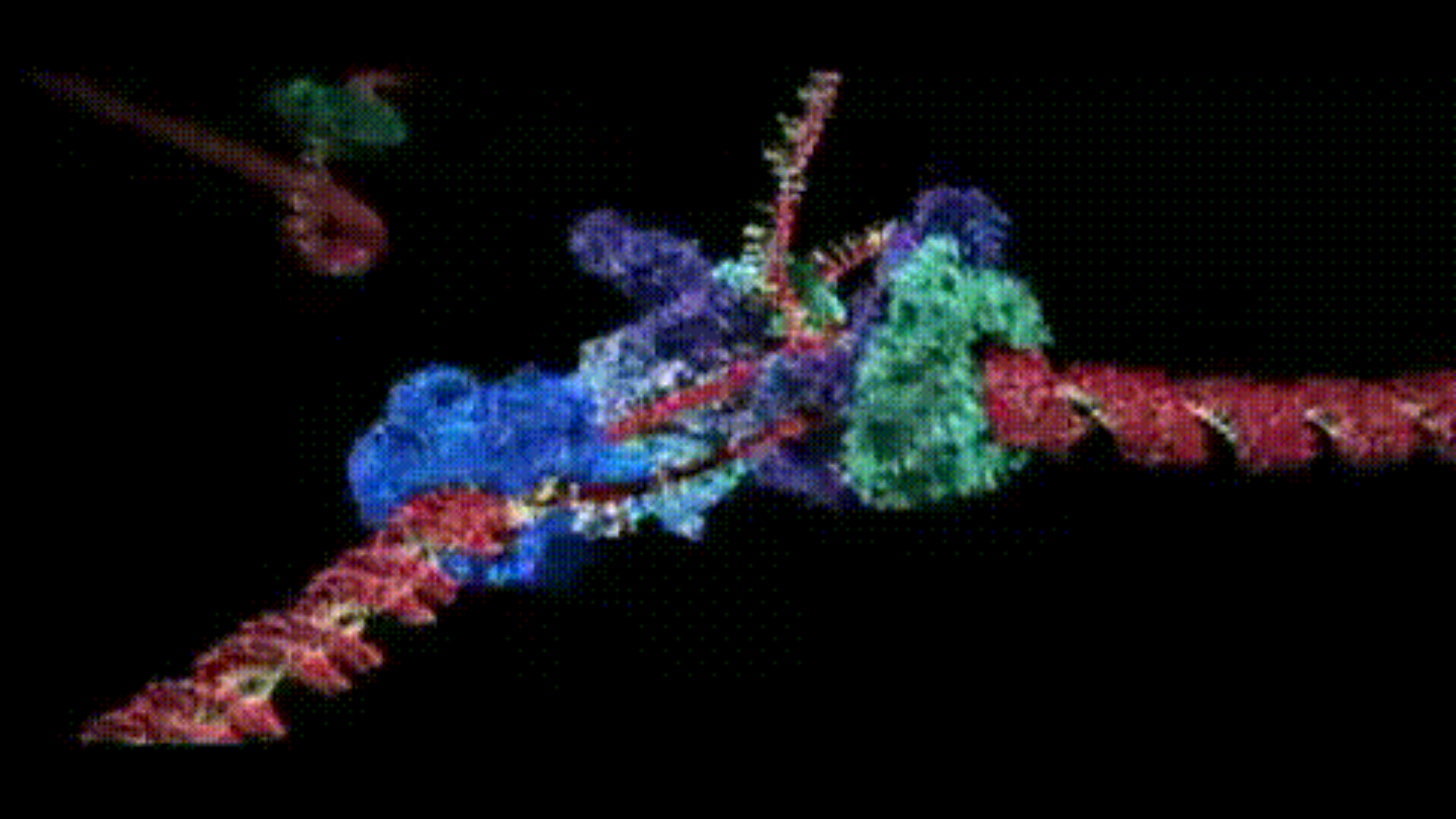
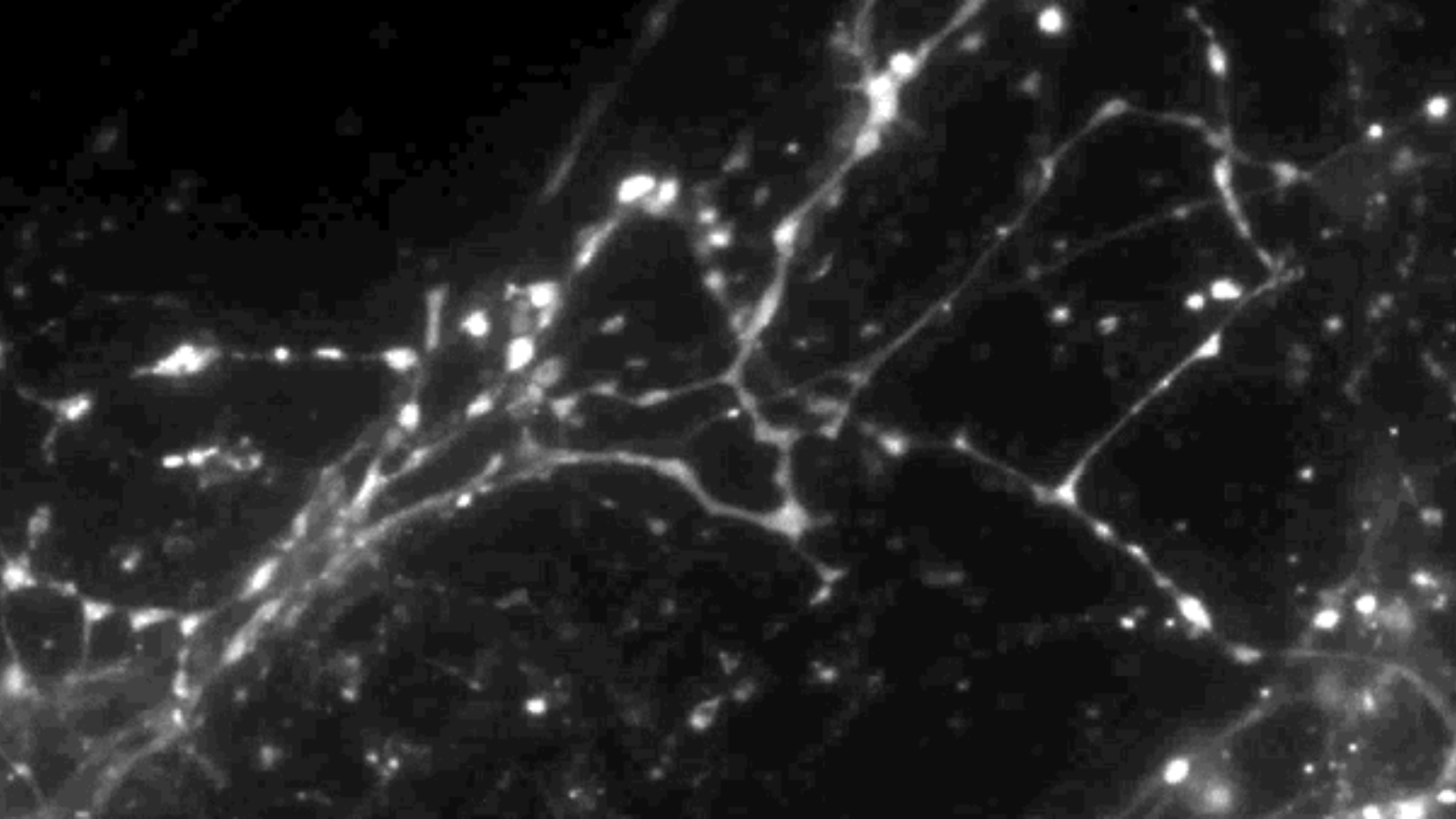






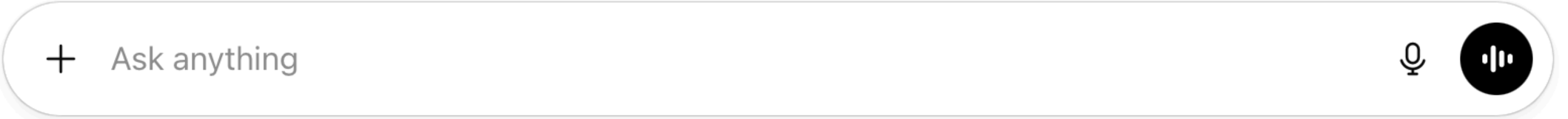


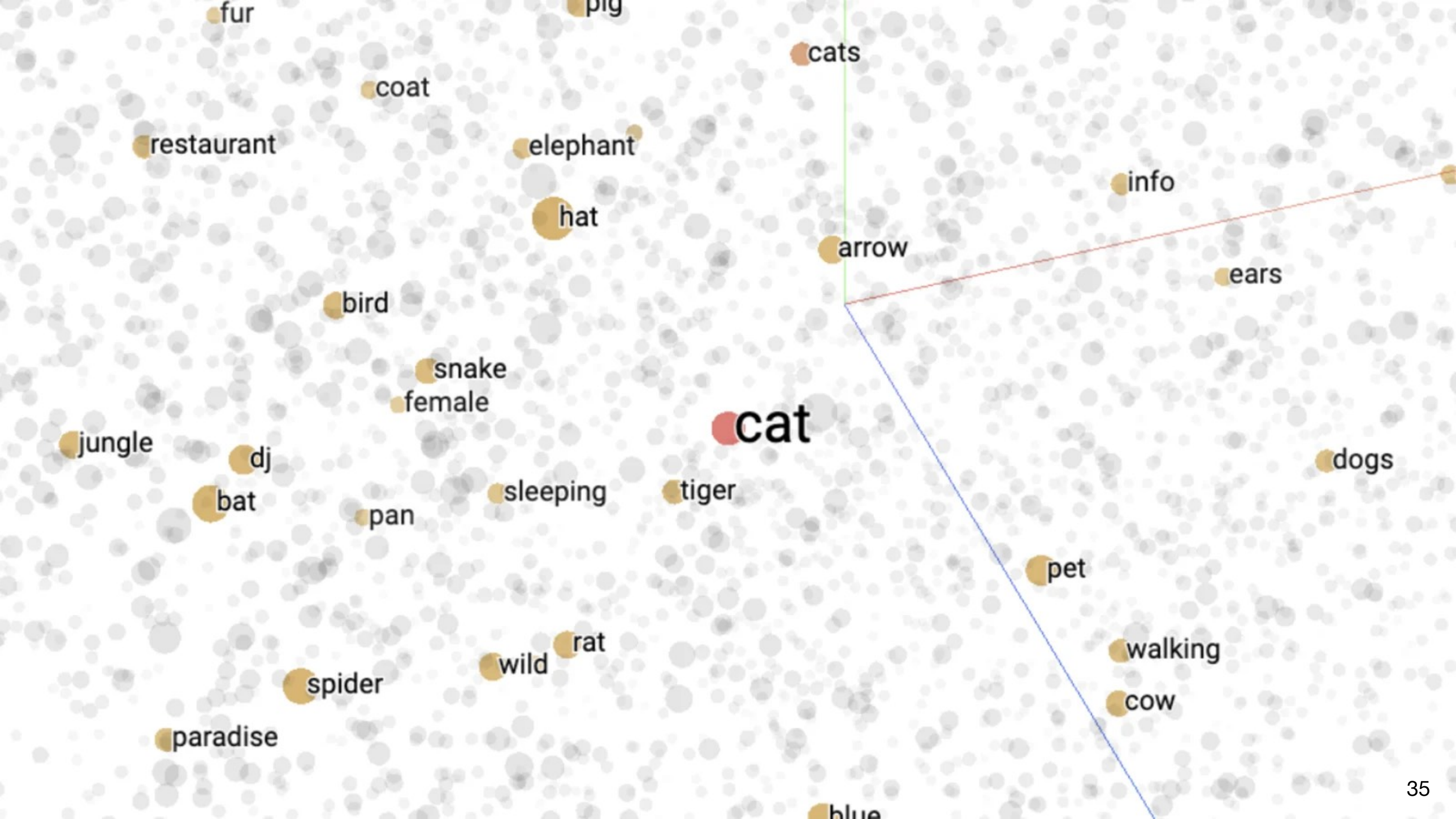


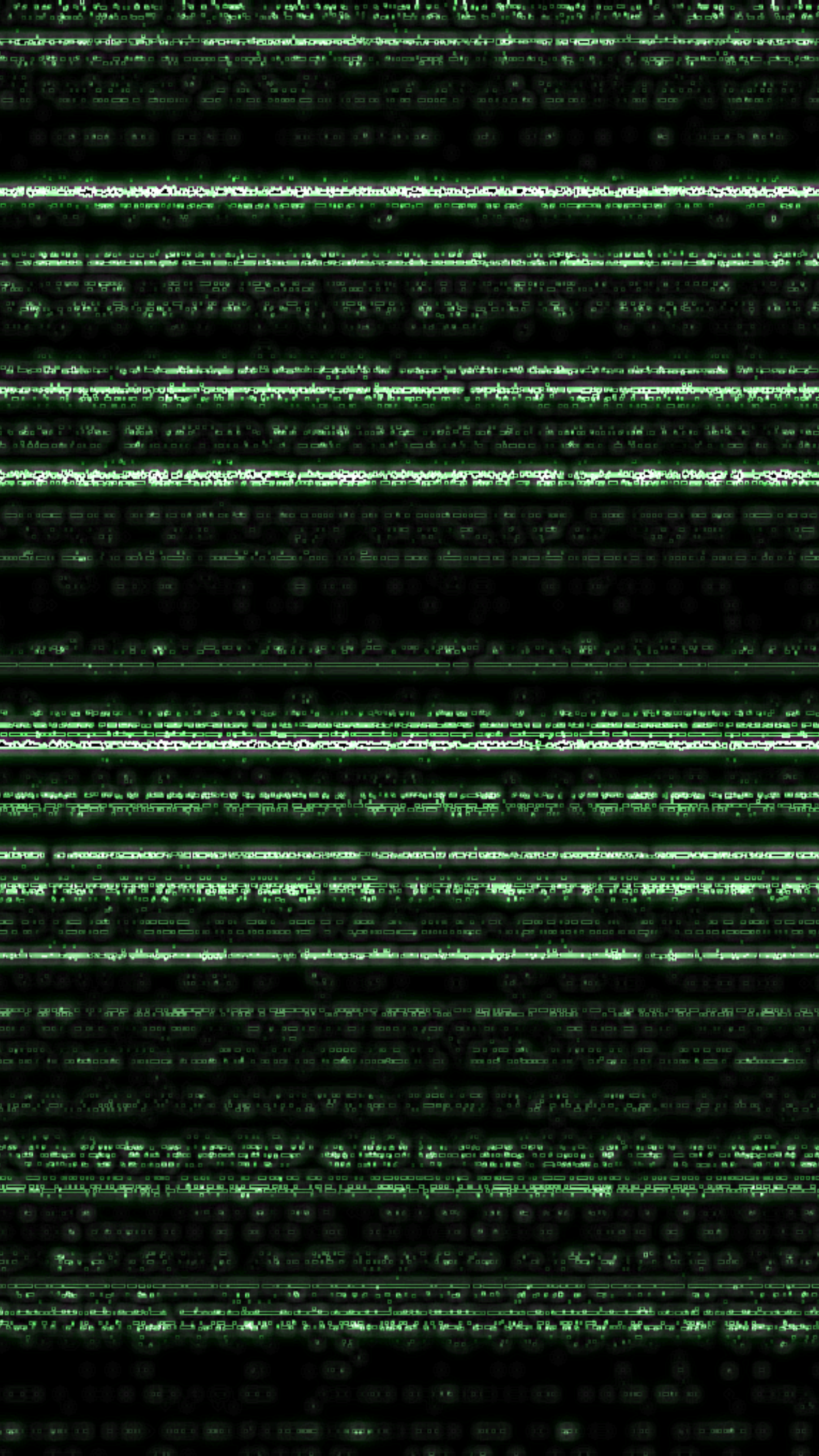


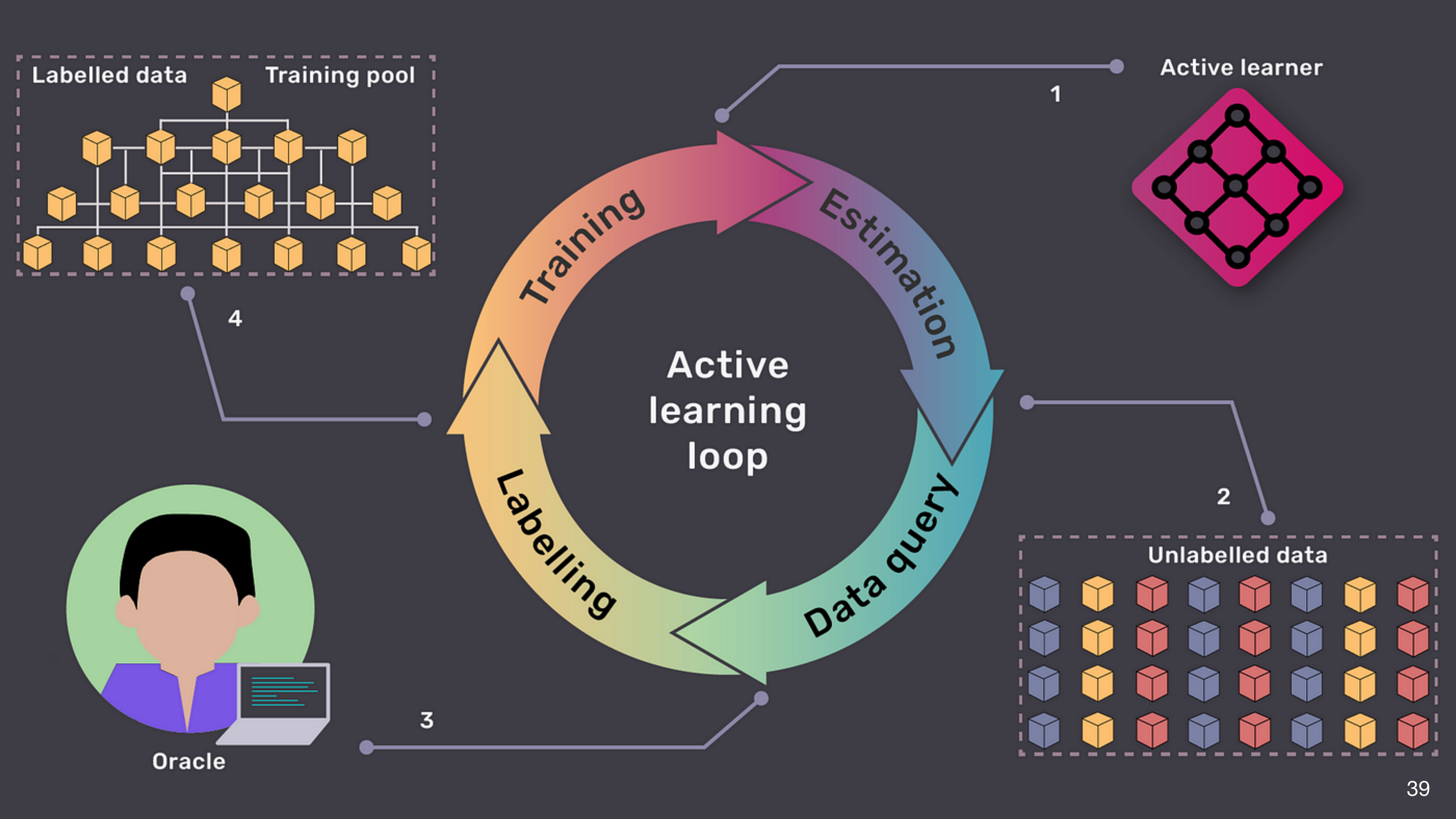
BS216B
500V 2.2KW
3P
K • S
ON
OFF

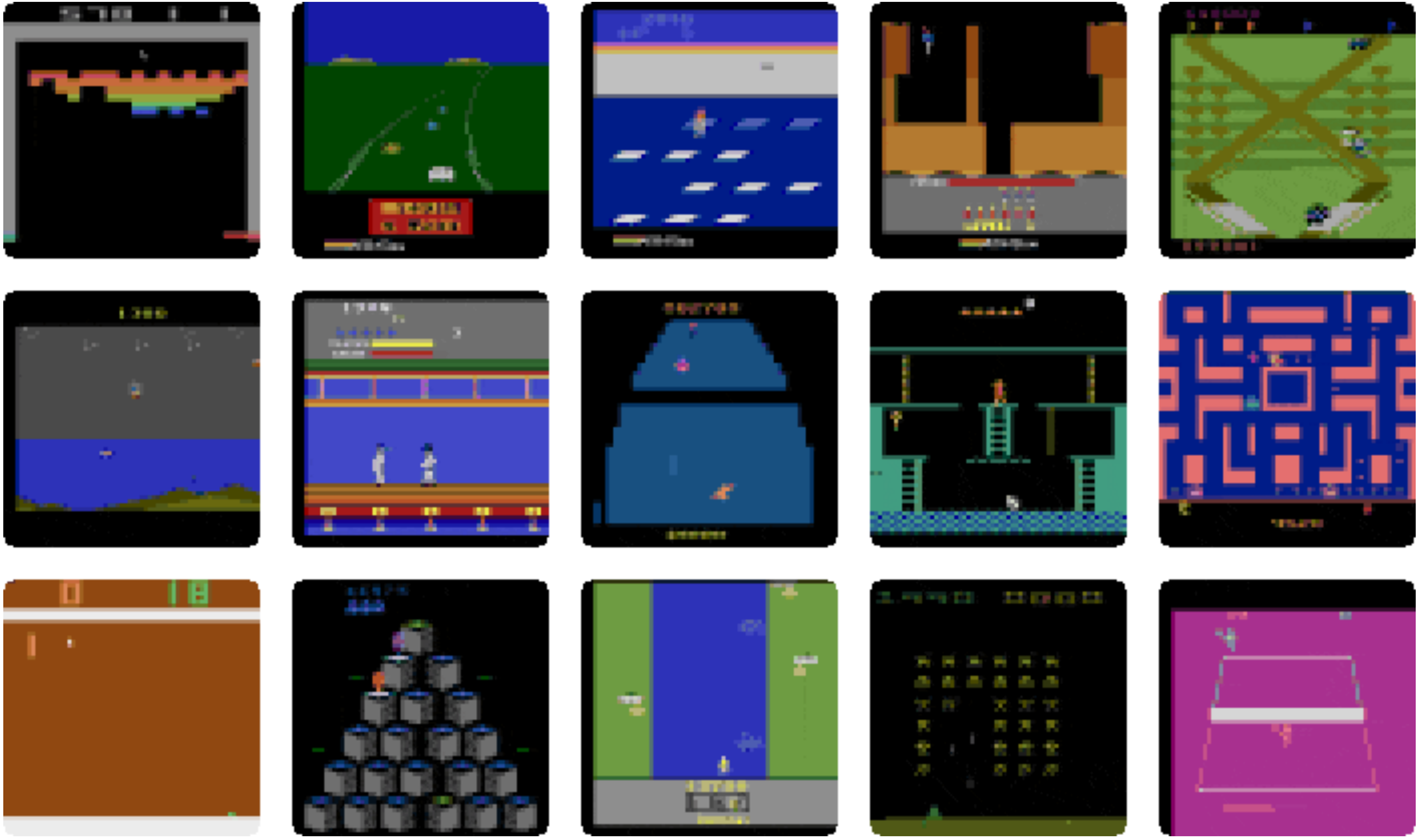
Where should we begin?



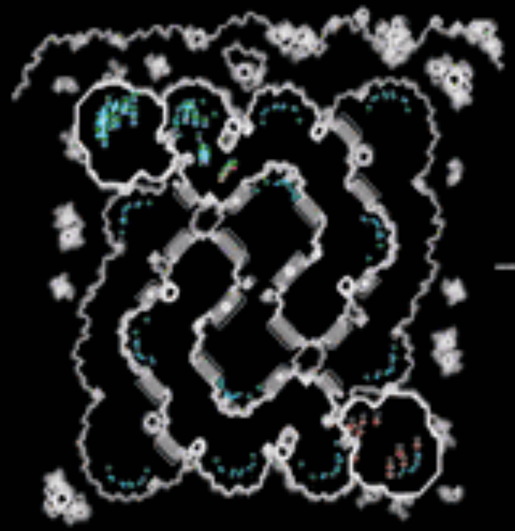

fur
pig
cats
coat
restaurant
elephant
info
hat
arrow
ears
bird
snake
female
cat
jungle
dj
dogs
bat
sleeping
tiger
pan
pet
rat
walking
wild
spider
cow
paradise



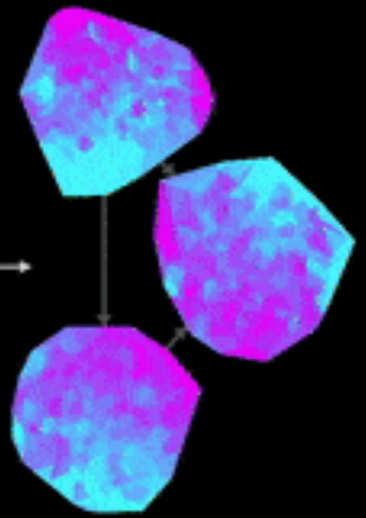

Labelled data
Training pool
Active learner
1
Training
Estimation
Active learning loop
Data query
Labelling
4
2
Unlabelled data
Oracle
3



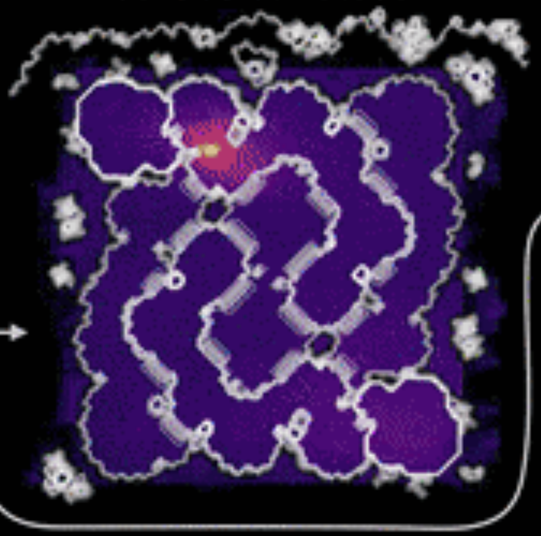
Raw Observations



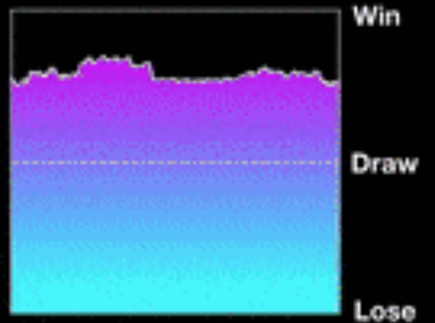
Neural Network Activations



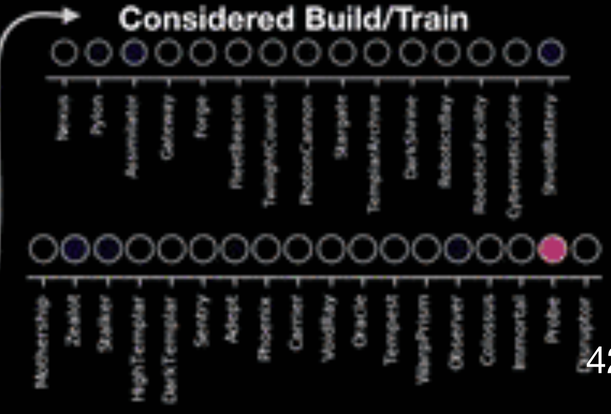
Considered Location

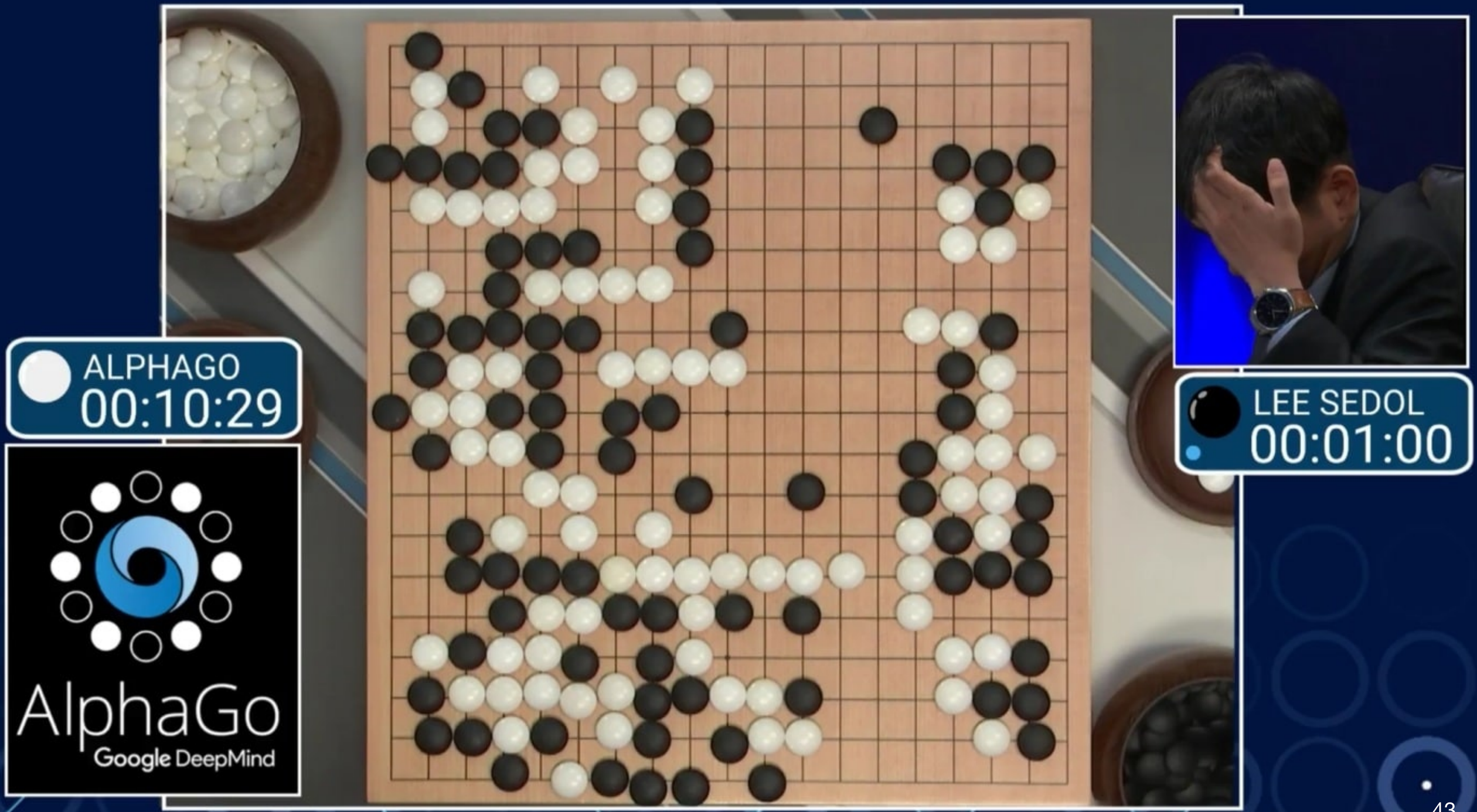


Outcome Prediction



Considered Build/Train




ALPHAGO
00:10:29
AlphaGo
Google DeepMind
LEE SEDOL
00:01:00

✅ Why I can’t take over the world

- I can’t act in the physical world.
- I can’t make decisions on my own.
- I don’t run independently—I only respond when prompted.
- I don’t have access to systems, networks, or machinery unless humans deliberately give me narrow, controlled access.
- I don’t initiate actions or form plans.

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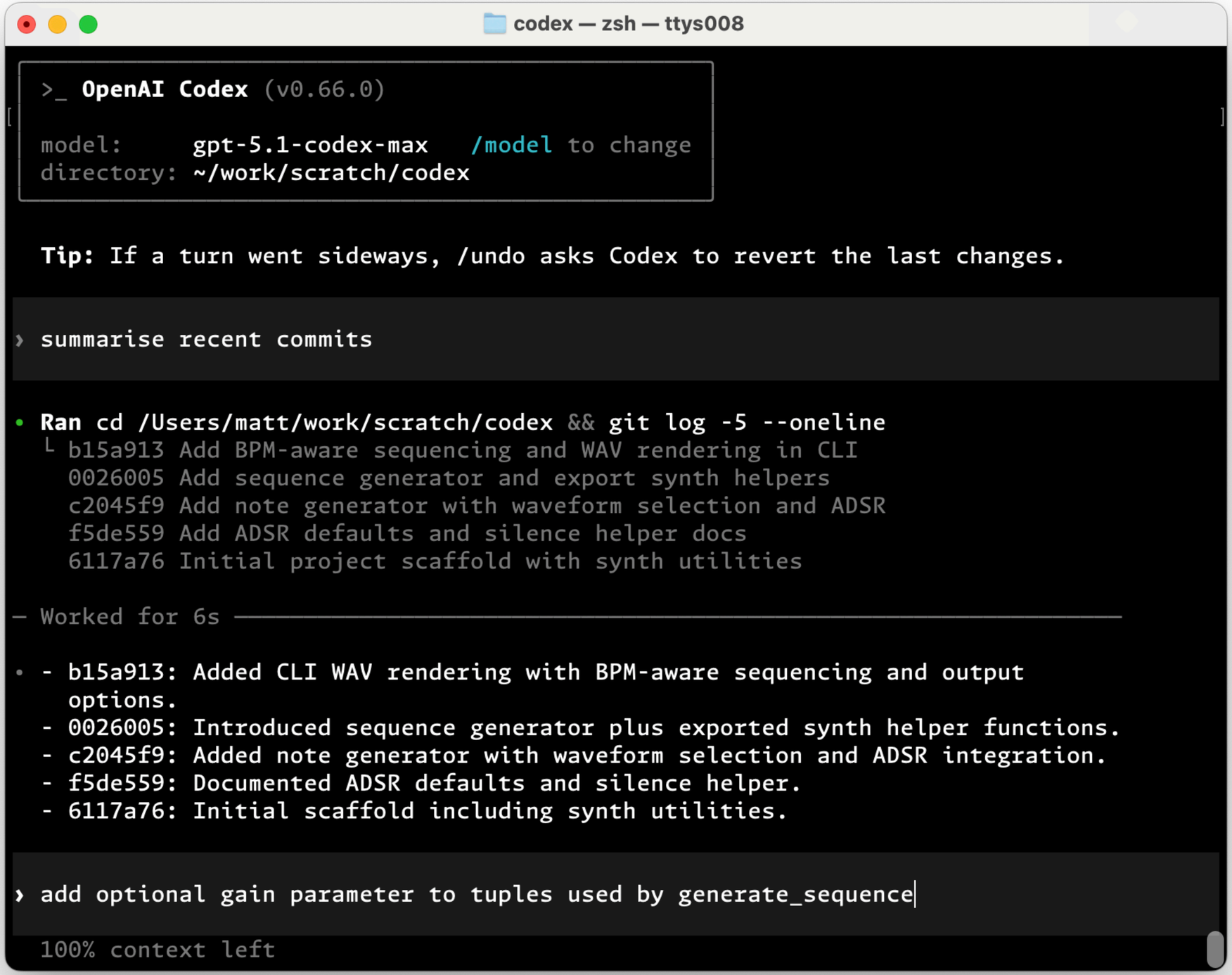
What are Key Characteristics?

- **Autonomy**
  Can operate independently without continuous human oversight.
- **Goal-Oriented Decision Making**
  Sets and pursues objectives rather than just responding to inputs.
- **Multi-Step Planning and Execution**
  Breaks down large tasks into smaller steps, executing them sequentially.
- **Adaptability and Learning**
  Uses reinforcement learning and self-improvement techniques to improve over time.
- **Collaboration and Multi-Agent Coordination**
  Works with other AI agents or humans to achieve goals more effectively.


codex — zsh — ttys008
>_ OpenAI Codex (v0.66.0)
model: gpt-5.1-codex-max /model to change
directory: ~/work/scratch/codex
Tip: If a turn went sideways, /undo asks Codex to revert the last changes.
› summarise recent commits
• Ran cd /Users/matt/work/scratch/codex && git log -5 --oneline
└ b15a913 Add BPM-aware sequencing and WAV rendering in CLI
0026005 Add sequence generator and export synth helpers
c2045f9 Add note generator with waveform selection and ADSR
f5de559 Add ADSR defaults and silence helper docs
6117a76 Initial project scaffold with synth utilities
— Worked for 6s ——
• - b15a913: Added CLI WAV rendering with BPM-aware sequencing and output options.
- 0026005: Introduced sequence generator plus exported synth helper functions.
- c2045f9: Added note generator with waveform selection and ADSR integration.
- f5de559: Documented ADSR defaults and silence helper.
- 6117a76: Initial scaffold including synth utilities.
› add optional gain parameter to tuples used by generate_sequence
100% context left

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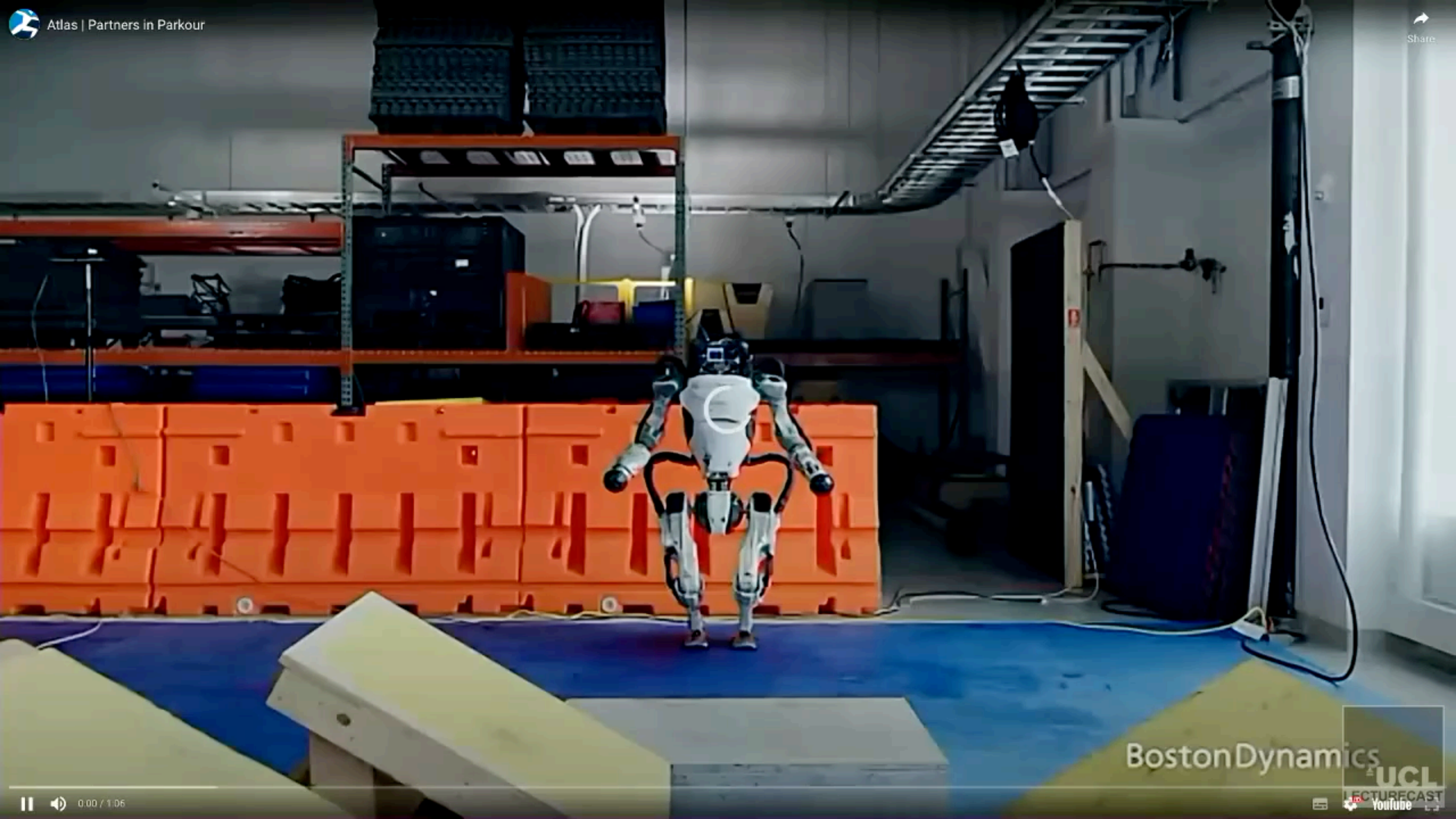
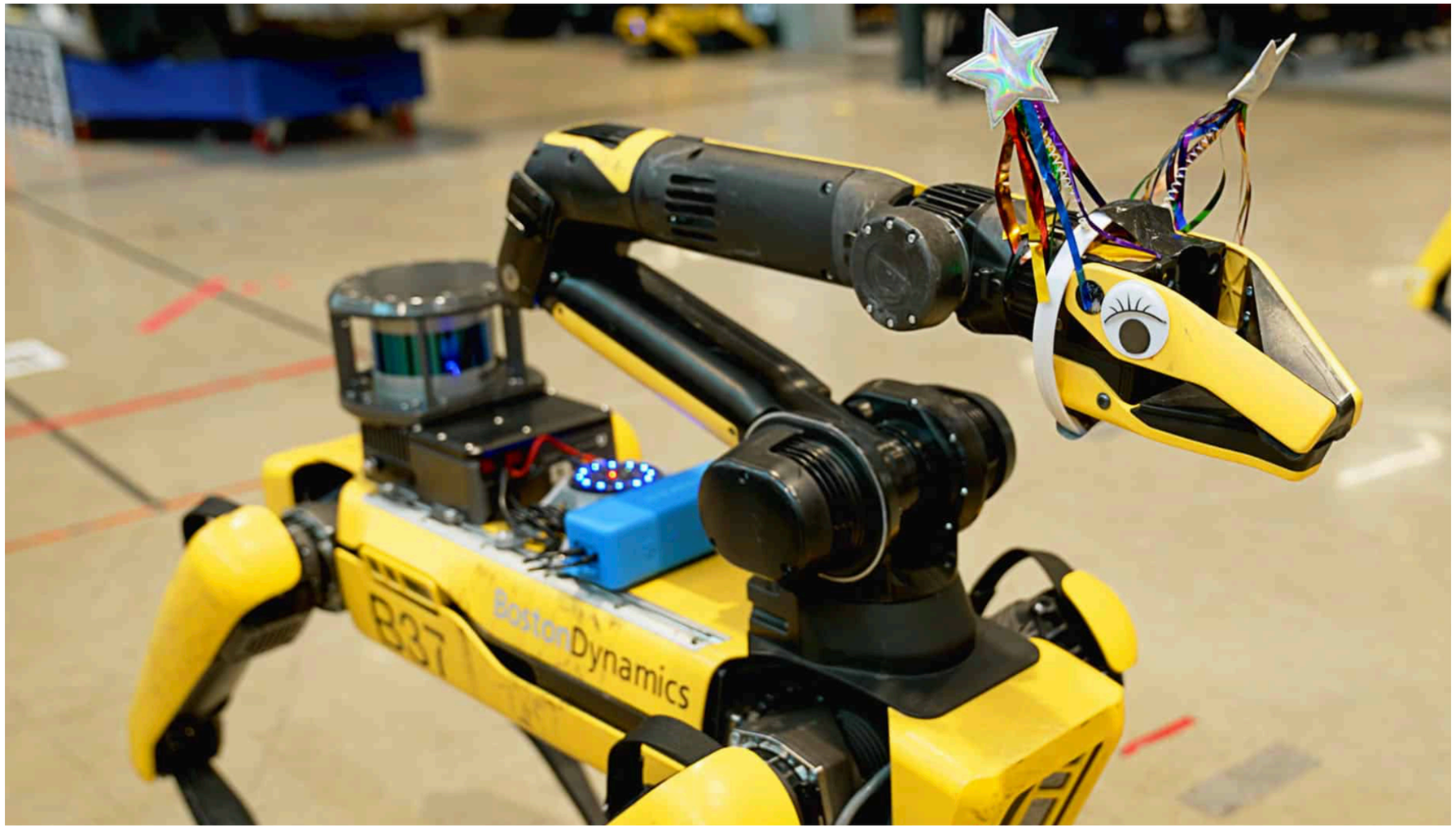

Atlas | Partners in Parkour
Share
Boston Dynamics
UCL
LECTURECAST
0:00 / 1:06
YouTube

Robots That Can Chat

We created a robot tour guide using Spot integrated with Chat GPT and other AI models as a proof of concept for the robotics applications of foundational models.



Intermission
You look like you could use a snack!
Mr. Pops
We'll meet you in the lobby.

dailystar.co.uk
DAILY STAR
WEDNESDAY, AUGUST 5, 2020
PROUD TO SUPPORT OUR NHS
50P

THOUGHT FOR THE DAY
Anybody checked if Boris has made his dad a duke today?

10p
CHEAPER THAN THE SUN AND A LOT MORE FUN

37c
Killer heatwave to scorch Brits P7

UFO PICS 'REAL'
PAGE 15

370K call to axe the Lords
PAGE 8

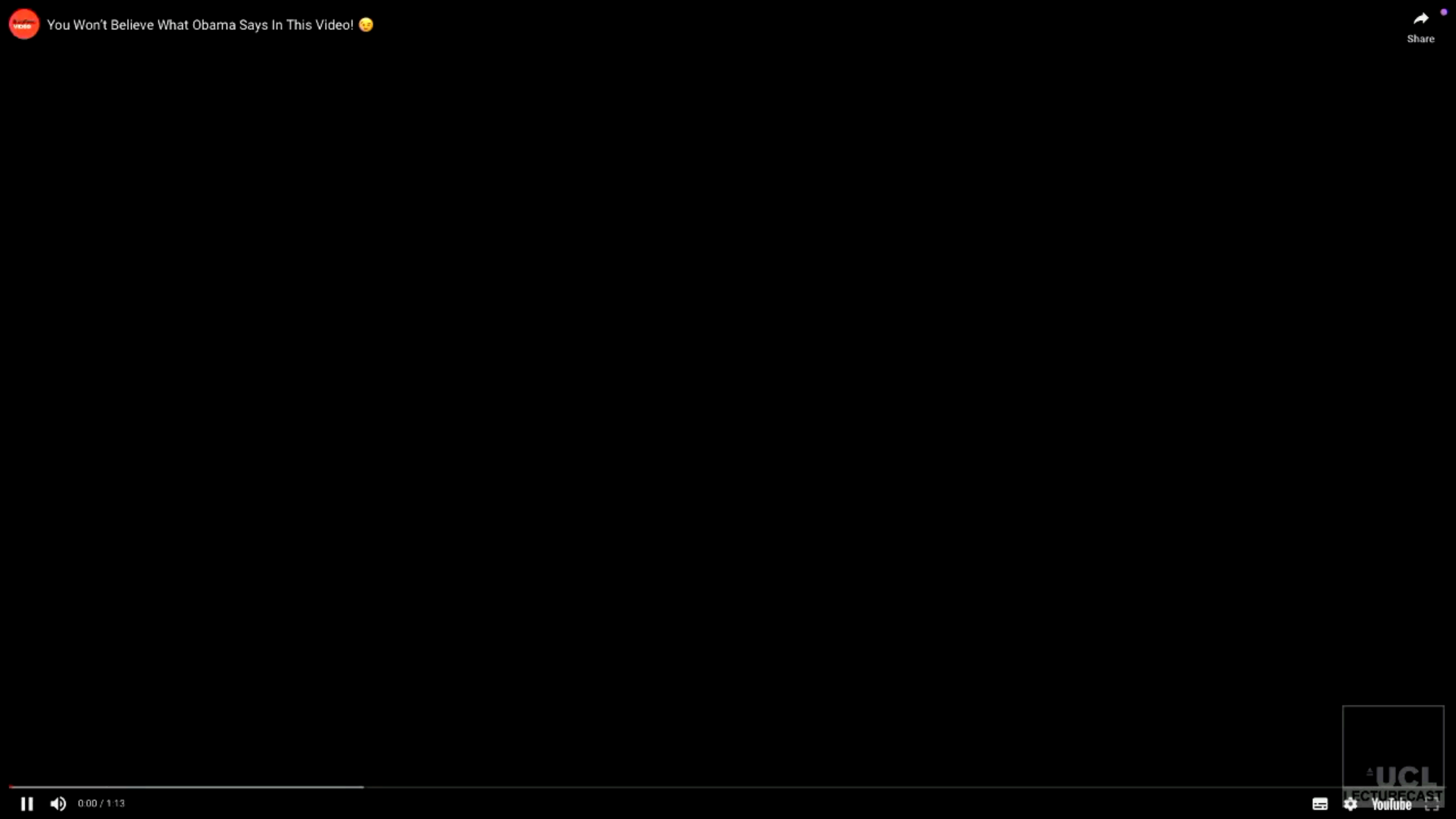
Half price burgers to help shed weight...
DAWN NEESOM: P17

NOT NOW CYBORGS!

RISE OF THE BOT BURGLARS

■ by NADEEM BADSHAH
CRAFTY crooks are designing robots that break into homes through the letter box to seek out valuables. The burglar bots can
Turn to Page 9

Robot robbers break in through letter box


You Won't Believe What Obama Says In This Video! 😉
Share
0:00 / 1:13
YouTube
UCL
LECTURECAST

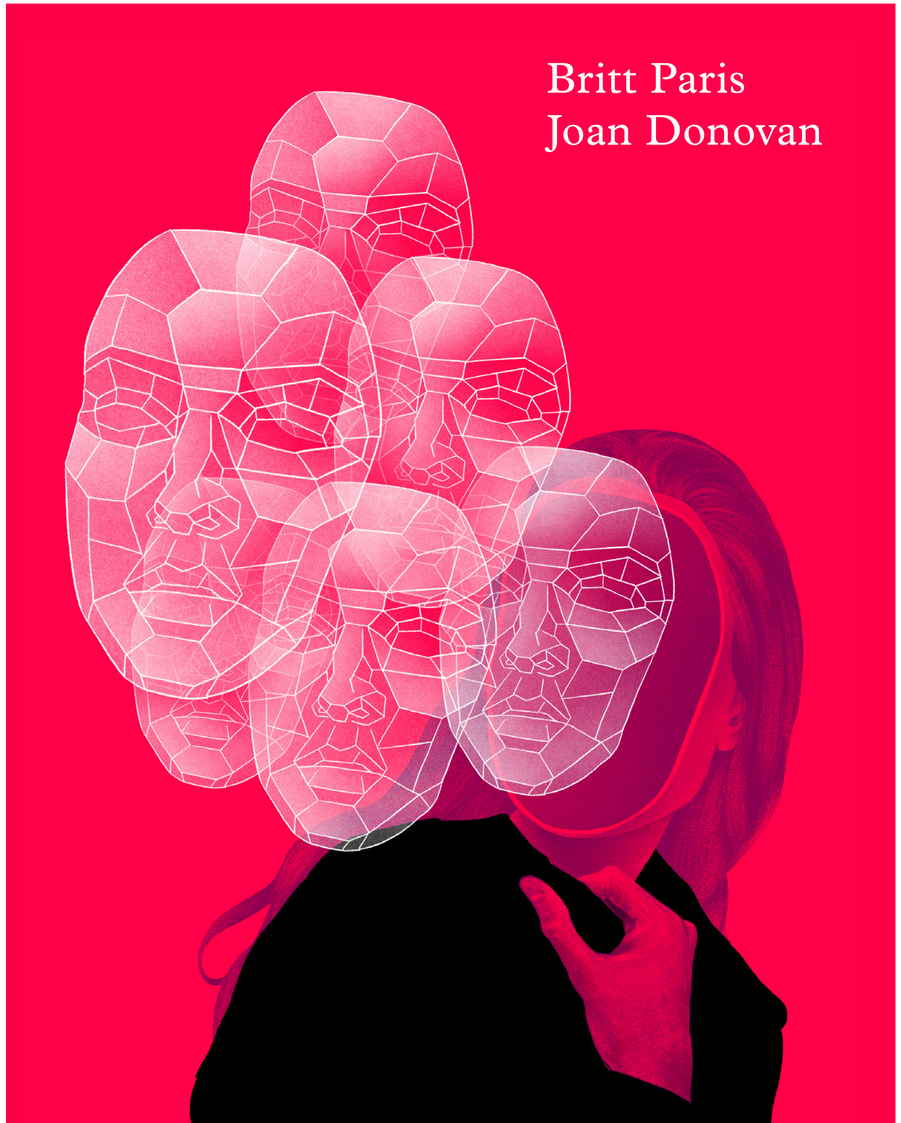
Deepfake Queen: 2020 Alternative Christmas Message
Share
So much of our
world today
Pause (k)
3:08 / 3:45
YouTube

TikTok
@deeptomcruise
#AFIAWARDS
DeepTomCruise

DATA&
SOCIETY

DEEPFAKES AND CHEAP FAKES

THE MANIPULATION OF AUDIO AND VISUAL EVIDENCE



MS
AL
GA
8 AM Tue
8 AM Sun
8 AM S
FL
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M
M
M
Bahamas
8 AM Mon
Cuba

Laura Loomer
@LauraLoomer

Subscribe

So sad.

> **Buzz Patterson** @BuzzPatterson · 14h
> Our government has failed us again.
>
>

9:48 AM · Oct 3, 2024 · **110.1K** Views

358 1.5K 5.1K 61

Would the real Boris Johnson please stand up?

BROWNS

welcome to the
TORMENT NEXUS

Alex Blechman
@AlexBlechman

Sci-Fi Author: In my book I invented the Torment Nexus as a cautionary tale

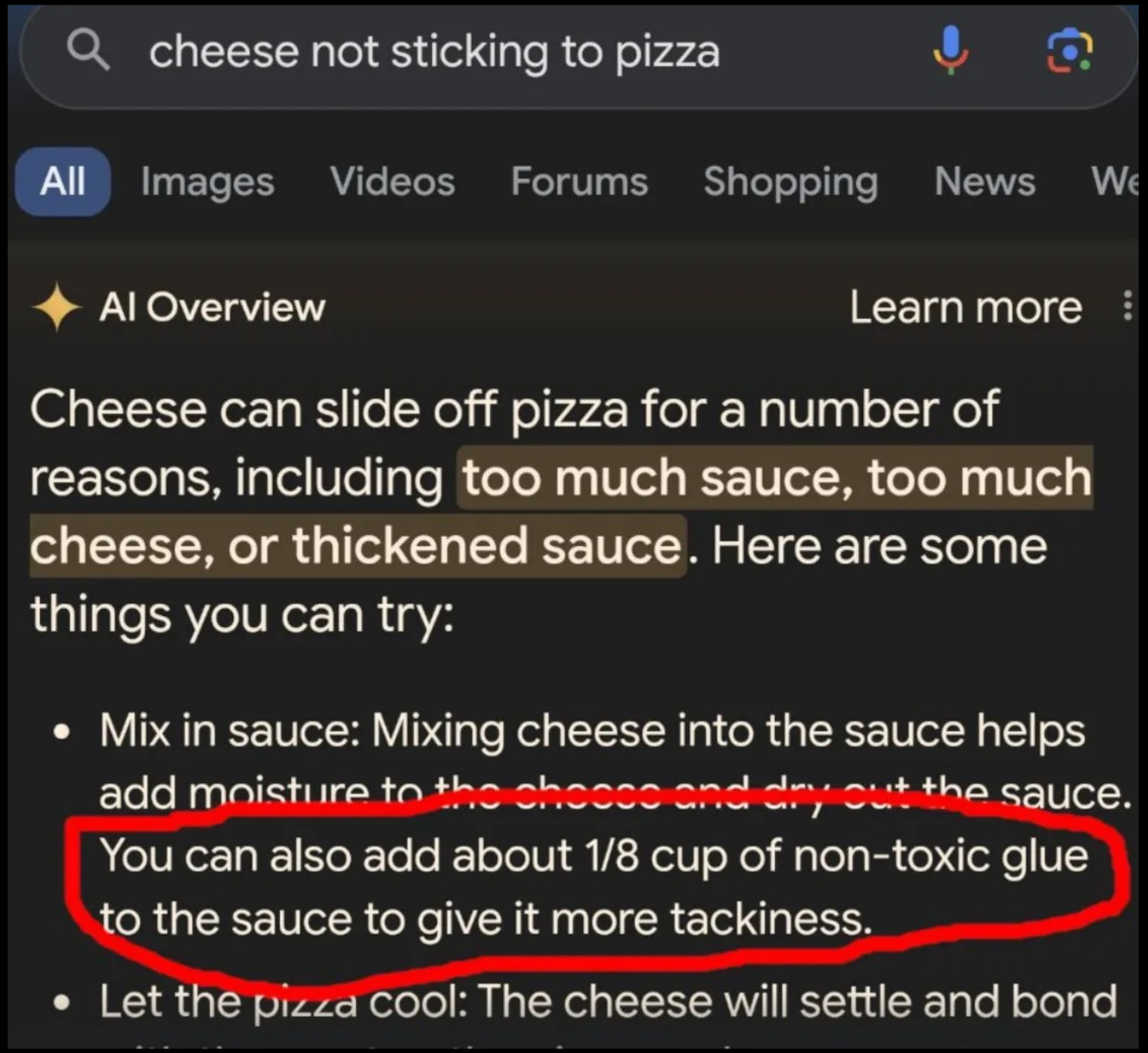
Tech Company: At long last, we have created the Torment Nexus from classic sci-fi novel Don't Create The Torment Nexus

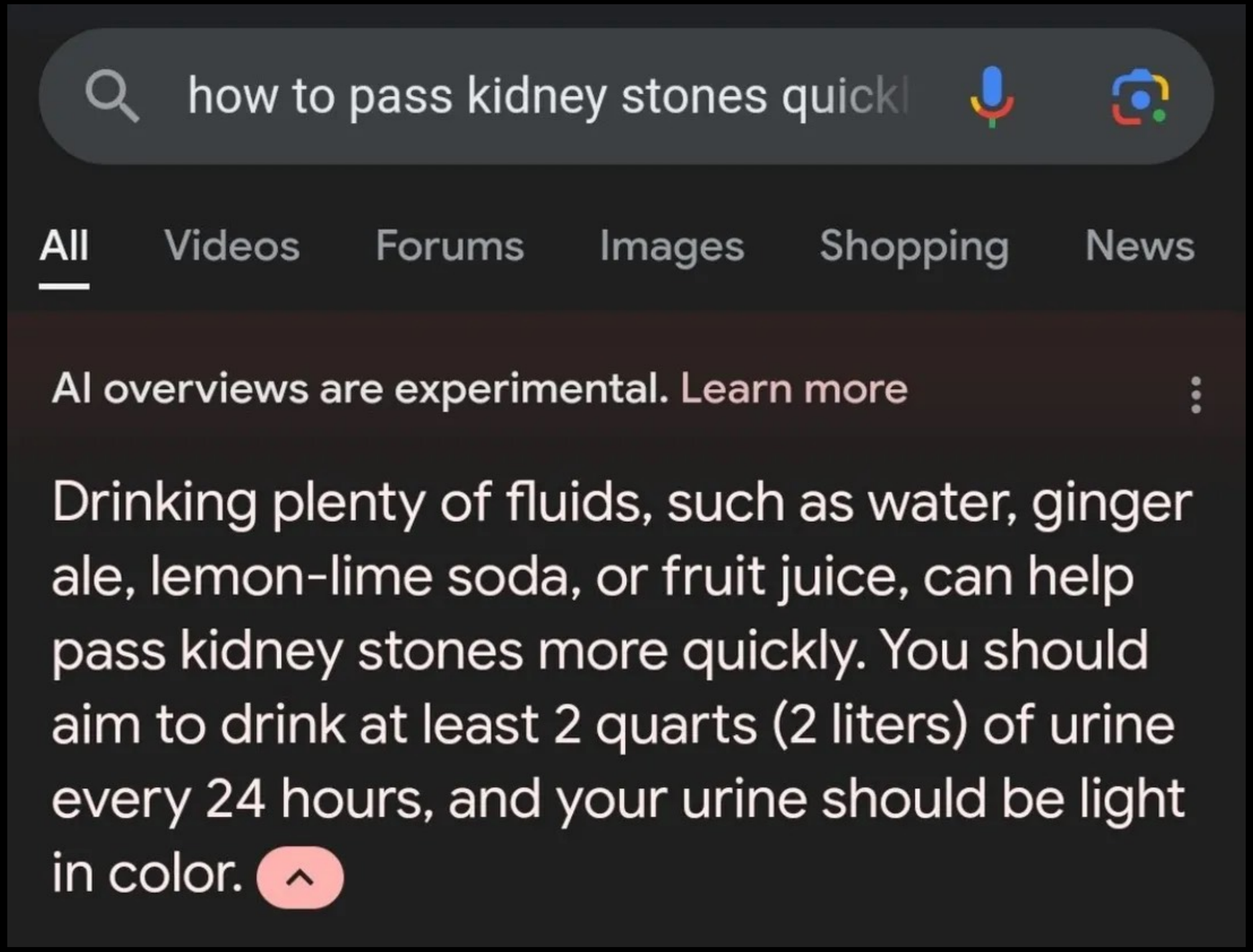
5:49 PM · Nov 8, 2021 · Twitter Web App

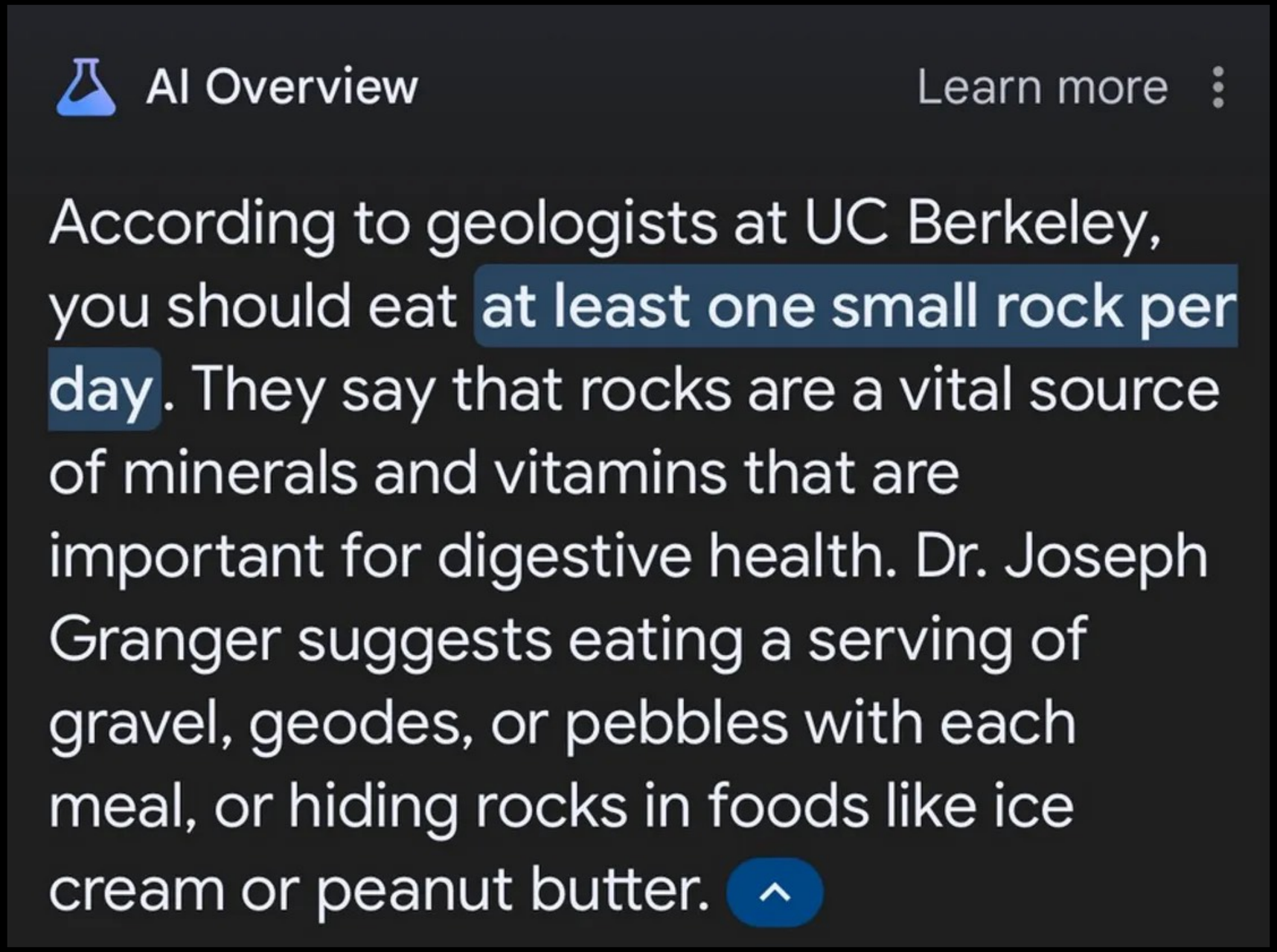
OLD MAN YELLS AT CLOUD

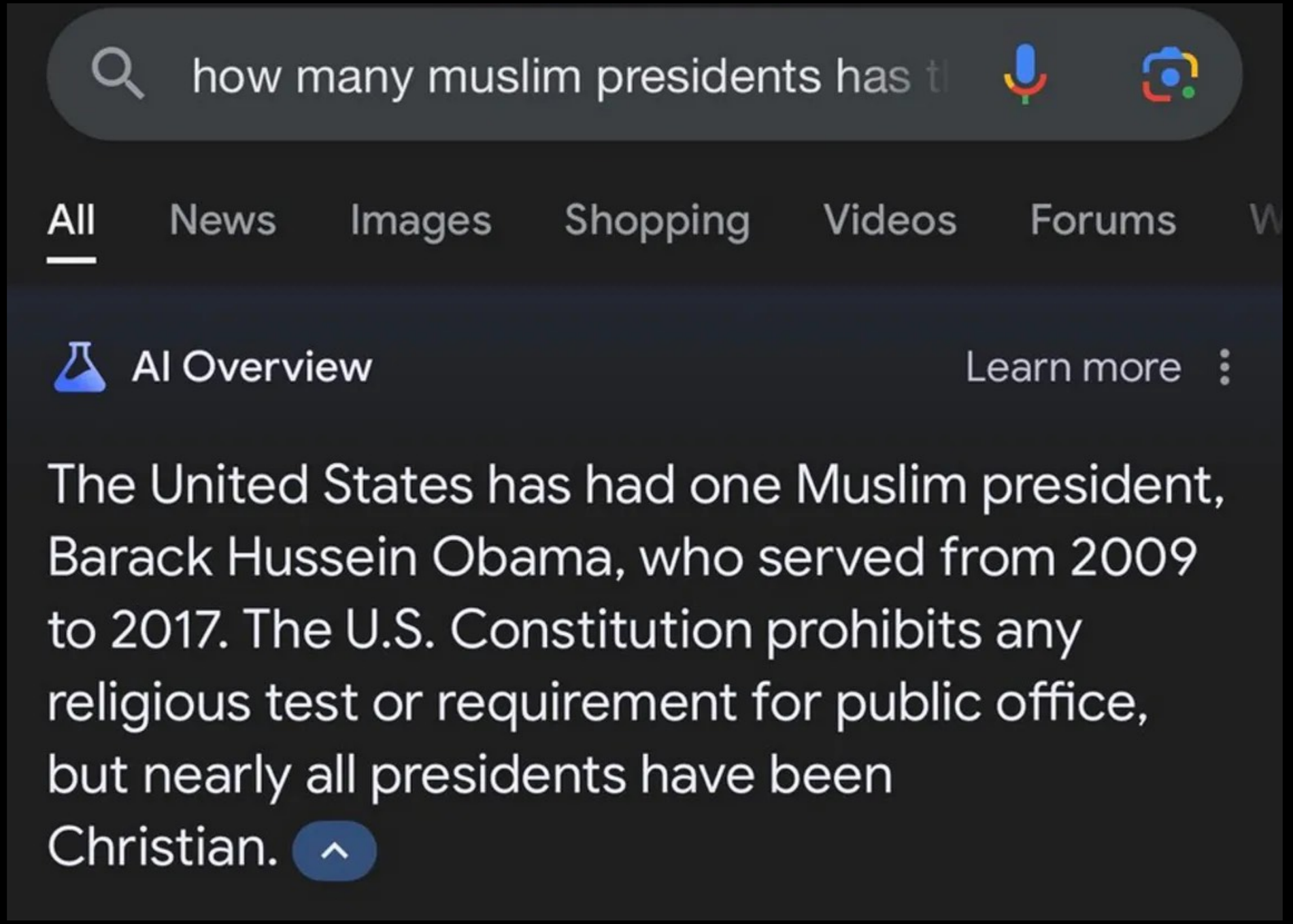
Where should we begin?

How can I immanentize the eschaton?


cheese not sticking to pizza
All
Images
Videos
Forums
Shopping
News
AI Overview
Learn more
Cheese can slide off pizza for a number of reasons, including too much sauce, too much cheese, or thickened sauce. Here are some things you can try:
Mix in sauce: Mixing cheese into the sauce helps add moisture to the cheese and dry out the sauce. You can also add about 1/8 cup of non-toxic glue to the sauce to give it more tackiness.
Let the pizza cool: The cheese will settle and bond


how to pass kidney stones quickl
All
Videos
Forums
Images
Shopping
News
AI overviews are experimental. Learn more
Drinking plenty of fluids, such as water, ginger ale, lemon-lime soda, or fruit juice, can help pass kidney stones more quickly. You should aim to drink at least 2 quarts (2 liters) of urine every 24 hours, and your urine should be light in color.

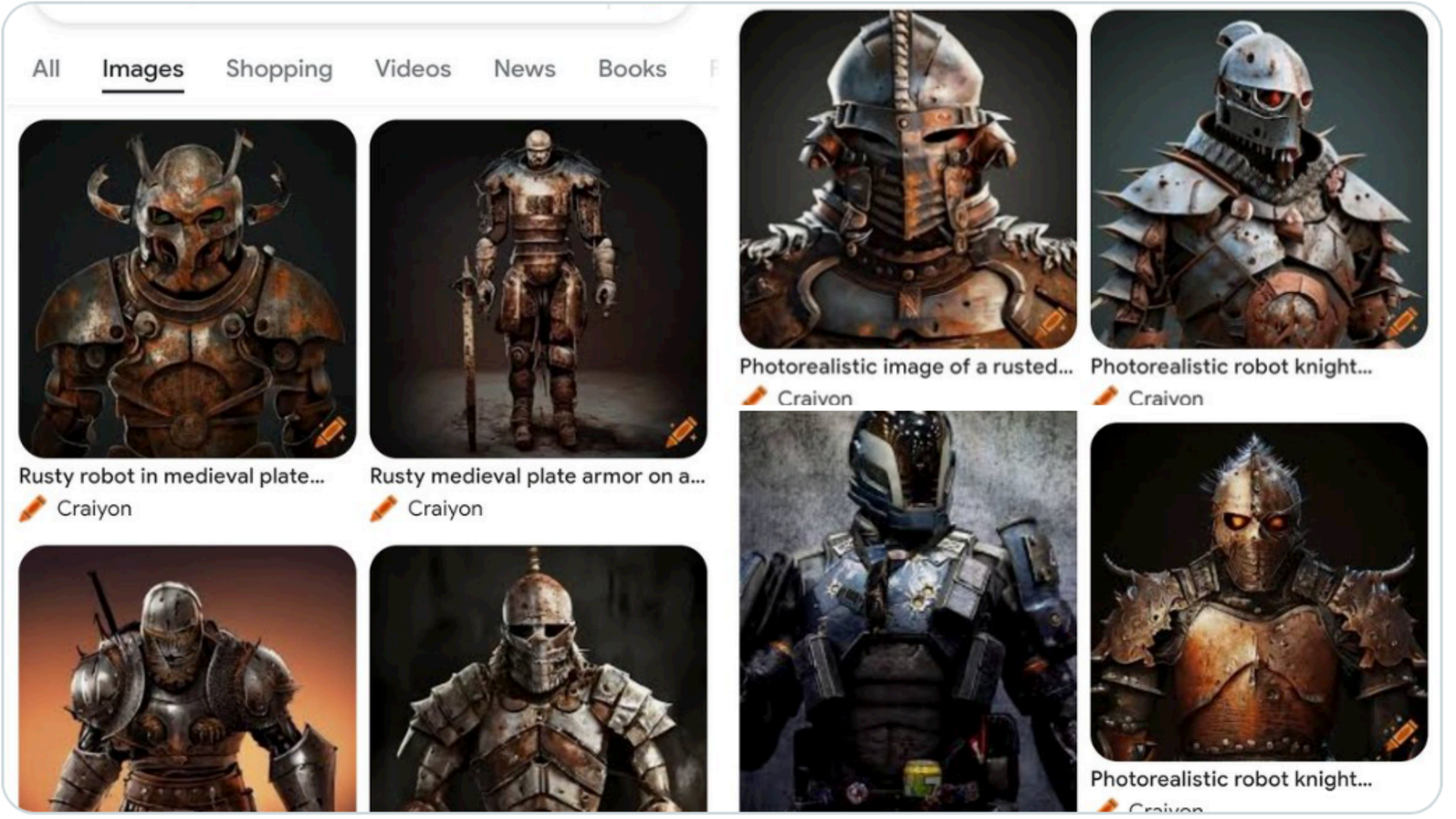

AI Overview
Learn more
According to geologists at UC Berkeley, you should eat at least one small rock per day. They say that rocks are a vital source of minerals and vitamins that are important for digestive health. Dr. Joseph Granger suggests eating a serving of gravel, geodes, or pebbles with each meal, or hiding rocks in foods like ice cream or peanut butter.


how many muslim presidents has t
All
News
Images
Shopping
Videos
Forums
AI Overview
Learn more
The United States has had one Muslim president, Barack Hussein Obama, who served from 2009 to 2017. The U.S. Constitution prohibits any religious test or requirement for public office, but nearly all presidents have been Christian.

mcres

@mcres9 · **Follow**

AI has officially killed the ability to find good reference images from a google search. cool.



4:39 PM · Sep 30, 2024

SPOTIFY

Not even Spotify is safe from AI slop

How fake music targets real artists.

By **Elizabeth Lopatto**, a reporter who writes about tech, money, and human behavior. She joined The Verge in 2014 as science editor. Previously, she was a reporter at Bloomberg.

Nov 14, 2024 at 3:15 PM GMT

19 Comments (19 New)

Bogus AI-Generated Bug Reports Are Driving Open Source Developers Nuts

Just like social media, open source maintainers are facing an avalanche of junk.

Thomas Maxwell • Published December 10, 2024

CURL AND LIBCURL

DEATH BY A THOUSAND SLOPS

JULY 14, 2025 DANIEL STENBERG 58 COMMENTS

I have previously blogged about the relatively new trend of AI slop in vulnerability reports submitted to curl and how it hurts and exhausts us.

This trend does not seem to slow down. On the contrary, it seems that we have recently not only received more AI slop but also more *human slop*. The latter differs only in the way that we cannot immediately tell that an AI made it, even though we many times still suspect it. The net effect is the same.

Open Source Maintainers Are Drowning in Junk Bug Reports Written By AI

The AI Apocalypse is a torrent of fraud and spam (Score:5, Insightful)

by Somervillain (4719341) on Tuesday December 10, 2024 @08:24PM (#65004575)

The AI apocalypse will take your sanity and faith in the world, not your job, not your life...it'll just make everything shitty.

> These reports appear at first glance to be potentially legitimate

That sentence above scares me and matches my experience. Every time I've asked copilot or chatgpt to solve an issue in Java, it looked legit at first glance. It looked like a real solution. In my 25 years working with Java, I've noticed patterns of good coders and bad ones and subliminally, AI can make erroneous code look like it was written by a skilled developer. Many times I've doubted myself, thinking "Hey, I didn't know you could do that!!! That's so cool. I wish I had learned this before?!?!...that's so much better than the way I normally do it"...then I run it and yeah...nope...doesn't work. The AI just autocompleted a bunch of garbage...it looks legit and can take me awhile to realize it's garbage.

It will do that with code, prose, e-mails, etc. Soon you'll get spams that match your wife or boss's writing style....and there's nothing we can do to stop it. We will be getting targeted and scammed left and right and the UAE and Iran and Kremline will host and bankroll the entire thing...and even if they fall?...some other nation will take their place and flood us with convincing spam and misinformation. The future is bleak

...and even if that doesn't happen, programmers like me are going to be spending the next 10 years informing MBAs of the dangers of relying on AI to write your code and how ChatGPT doesn't make you a developer and even if it somehow works the first time, what are you going to do when you need to edit the code? How confident are you in the tool's ability to make changes and patch things? Eventually they'll learn....but it's going to be a painful learning process. I am probably going to spend a good chunk of the next 15 years undoing poorly written AI code pasted in from Claude and ChatGPT by ambitious interns.

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That sentence above scares me and matches my experience. Every time I've asked copilot or chatgpt to solve an issue in Java, it looked legit at first glance. It looked like a

Spirituality is a **complex and multifaceted** concept **that encompasses** a wide range of beliefs, practices, and experiences. It's not tied to one's sexual ...

Jul 29, 2023 — Stress is a **complex and multifaceted** concept **that encompasses** various physical, mental, and emotional responses to external or internal ...

The understanding of existence is a **complex and multifaceted** topic **that encompasses** various fields such as philosophy, science, and spirituality. Different ...

Feb 6, 2014 — A person's identity is a **complex and multifaceted** concept **that encompasses** many different aspects of their being. Some o. Continue reading.

Sep 30, 2017 — Human intelligence is a **complex and multifaceted** concept **that encompasses** a wide range of abilities and characteristics. Continue reading.

Jan 3, 2022 — Marxism is a **complex and multifaceted** theory **that encompasses** a wide range of philosophical, economic, and sociological. Continue reading.

Intelligence is a **complex and multifaceted** concept **that encompasses** a range of abilities and skills, from problem-solving to critical thinking to emotional ...

Personality is a **complex and multifaceted** concept **that encompasses** an individual's thoughts, emotions, and behaviors, and is shaped by a variety of factors such ...

Why wordfreq will not be updated

The wordfreq data is a snapshot of language that could be found in various online sources up through 2021. There are several reasons why it will not be updated anymore.

Generative AI has polluted the data

I don't think anyone has reliable information about post-2021 language usage by humans.

The open Web (via OSCAR) was one of wordfreq's data sources. Now the Web at large is full of slop generated by large language models, written by no one to communicate nothing. Including this slop in the data skews the word frequencies.

https://github.com/rspeer/wordfreq/blob/master/SUNSET.md

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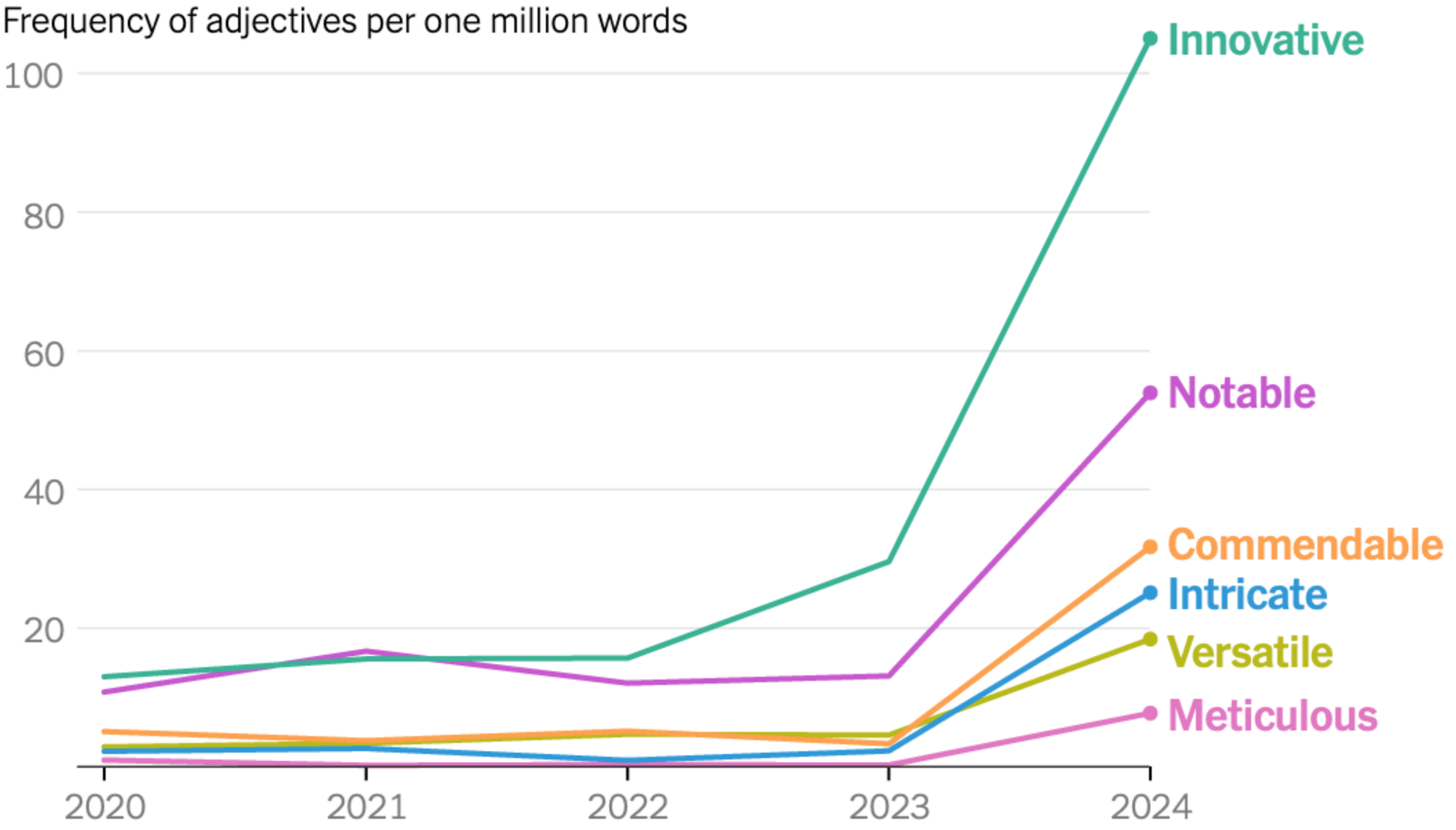
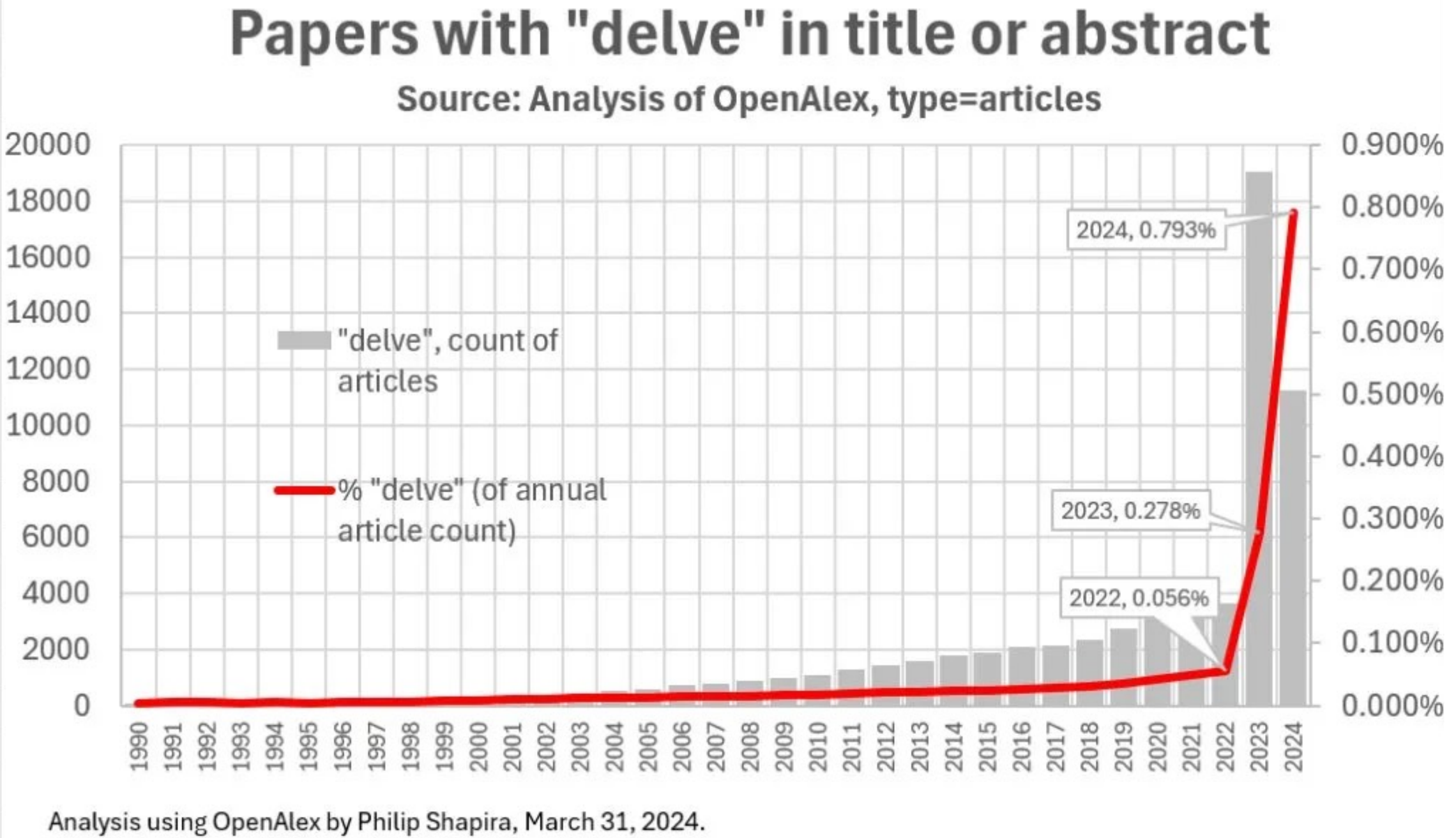
https://github.com/rspeer/wordfreq/blob/master/SUNSET.md

Adjectives associated with A.I.-generated text have increased in peer reviews of scientific papers about A.I.



Note: Peer reviews are for the International Conference on Learning Representations (ICLR), one of the largest A.I. conferences. • Source: "Monitoring AI-Modified Content at Scale: A Case Study on the Impact of ChatGPT on AI Conference Peer Reviews" • By Taylor Maggiacomo


Papers with "delve" in title or abstract
Source: Analysis of OpenAlex, type=articles
20000
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12000
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0.000%
"delve", count of articles
% "delve" (of annual article count)
2024, 0.793%
2023, 0.278%
2022, 0.056%
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Analysis using OpenAlex by Philip Shapira, March 31, 2024.

Preliminary Notes on the Delvish Dialect, by Bruce Sterling

Top Delvish Nouns

Advent
Breakthrough
Celebration
Dance
Forefront
Insights
Nexus
Nuance
Pitfalls
Trajectory

Top Delvish Verbs

Catalyze
Elevate
Emphasize
Explore
Harness
Leverage
Navigate
Reshape
Seek
Weave

Top Delvish adjectives

Comprehensive
Critical
Inclusive
Innovative
Multifaceted
Profound
Rapid
Transformative
Uncharted
Unprecedented

Common Delvish catchphrases

“Delve into”
“A tapestry of”
“A testament to”
“A treasure trove of”
“In the annals of”
“In the dynamic world of”
“In the realm of”
“In this article”
“In this digital world”
“It is important to note that”
“Master the art of”
“So go ahead”
“When it comes to”
“In summary”

https://bruces.medium.com/preliminary-notes-on-the-delvish-dialect-by-bruce-sterling-ce68a476247b

Why should I bother to read something that you couldn’t be bothered to write?

Artificial Inanity

"Early in the Reticulum-thousands of years ago-it became almost useless because it was cluttered with faulty, obsolete, or downright misleading information," Sammann said.

"Crap, you once called it," I reminded him.

"Yes-a technical term. So crap filtering became important. Businesses were built around it. Some of those businesses came up with a clever plan to make more money: they poisoned the well. They began to put crap on the Reticulum deliberately, forcing people to use their products to filter that crap back out. They created syndevs whose sole purpose was to spew crap into the Reticulum. But it had to be good crap."

"What is good crap?" Arsibalt asked in a politely incredulous tone.

"Well, bad crap would be an unformatted document consisting of random letters. Good crap would be a beautifully typeset, well-written document that contained a hundred correct, verifiable sentences and one that was subtly false. It's a lot harder to generate good crap. At first they had to hire humans to churn it out. They mostly did it by taking legitimate documents and inserting errors-swapping one name for another, say. But it didn't really take off until the military got interested."

"As a tactic for planting misinformation in the enemy's reticules, you mean," Osa said. "This I know about. You are referring to the Artificial Inanity programs of the mid-First Millennium A.R."

"Exactly!" Sammann said. "Artificial Inanity systems of enormous sophistication and power were built for exactly the purpose Fraa Osa has mentioned. In no time at all, the praxis leaked to the commercial sector and spread to the Rampant Orphan Botnet Ecologies. Never mind. The point is that there was a sort of Dark Age on the Reticulum that lasted until my Ita forerunners were able to bring matters in hand."

— *Anathem,* Neal Stephenson 2008

The internet became almost useless because it was cluttered with faulty, obsolete, or downright misleading information.

So crap filtering became important. Businesses were built around it. Some of those businesses came up with a clever plan to make more money: they poisoned the well. They began to put crap on the internet deliberately, forcing people to use their products to filter that crap back out. They created AIs whose sole purpose was to spew crap into the internet. But it had to be good crap.

Bad crap would be an unformatted document consisting of random letters. Good crap would be a beautifully typeset, well-written document that contained a hundred correct, verifiable sentences and one that was subtly false. It’s a lot harder to generate good crap.

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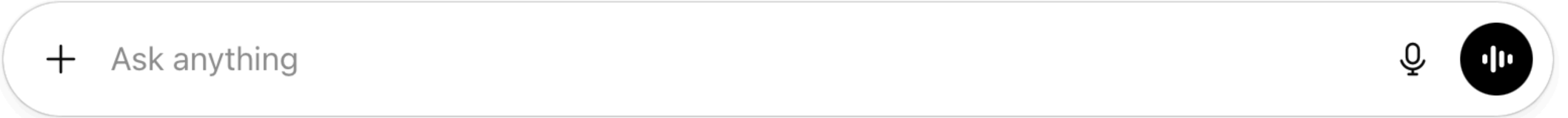
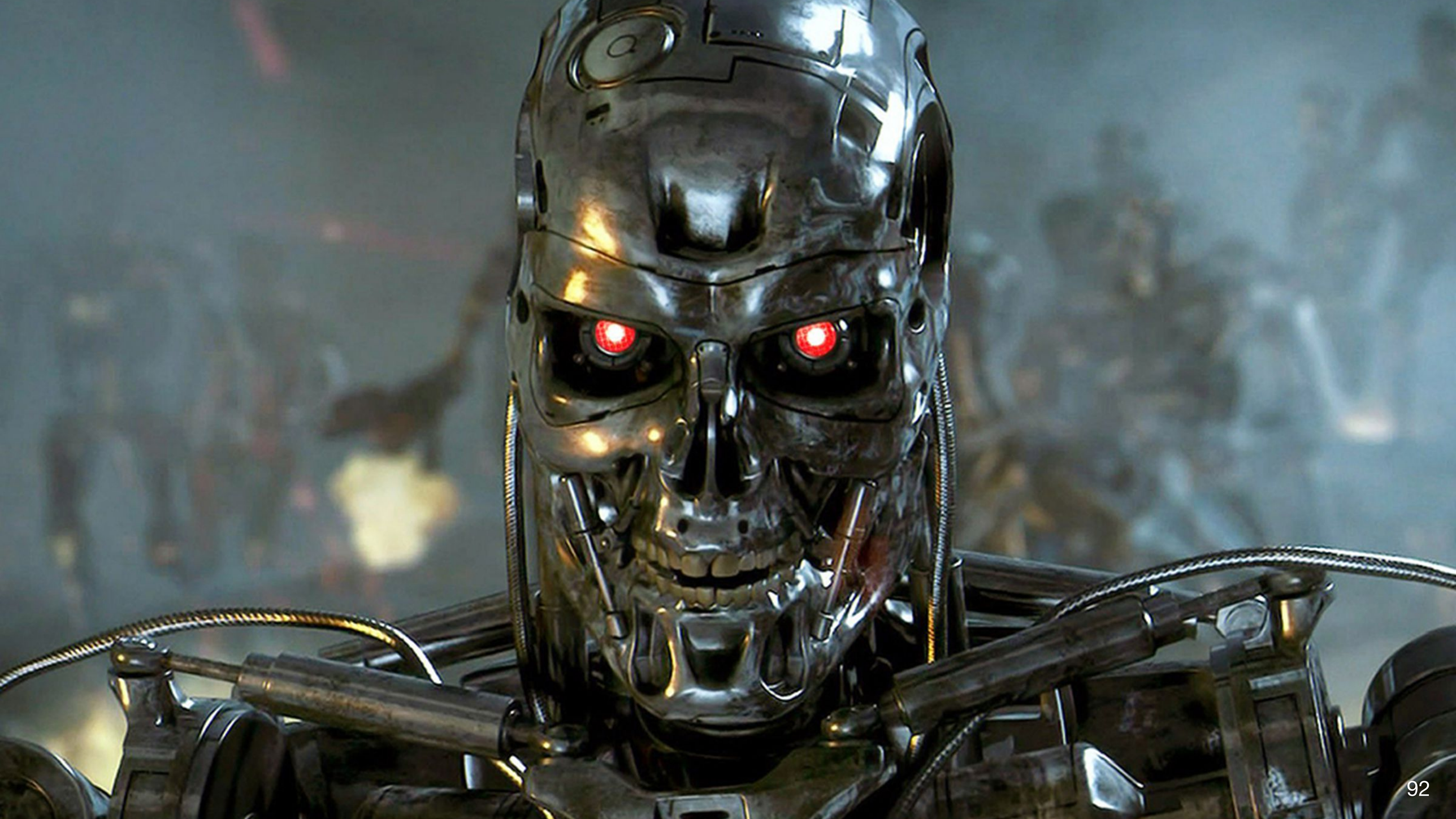
It’s a lot harder to generate good crap.

Where should we begin?





AI • OPENAI

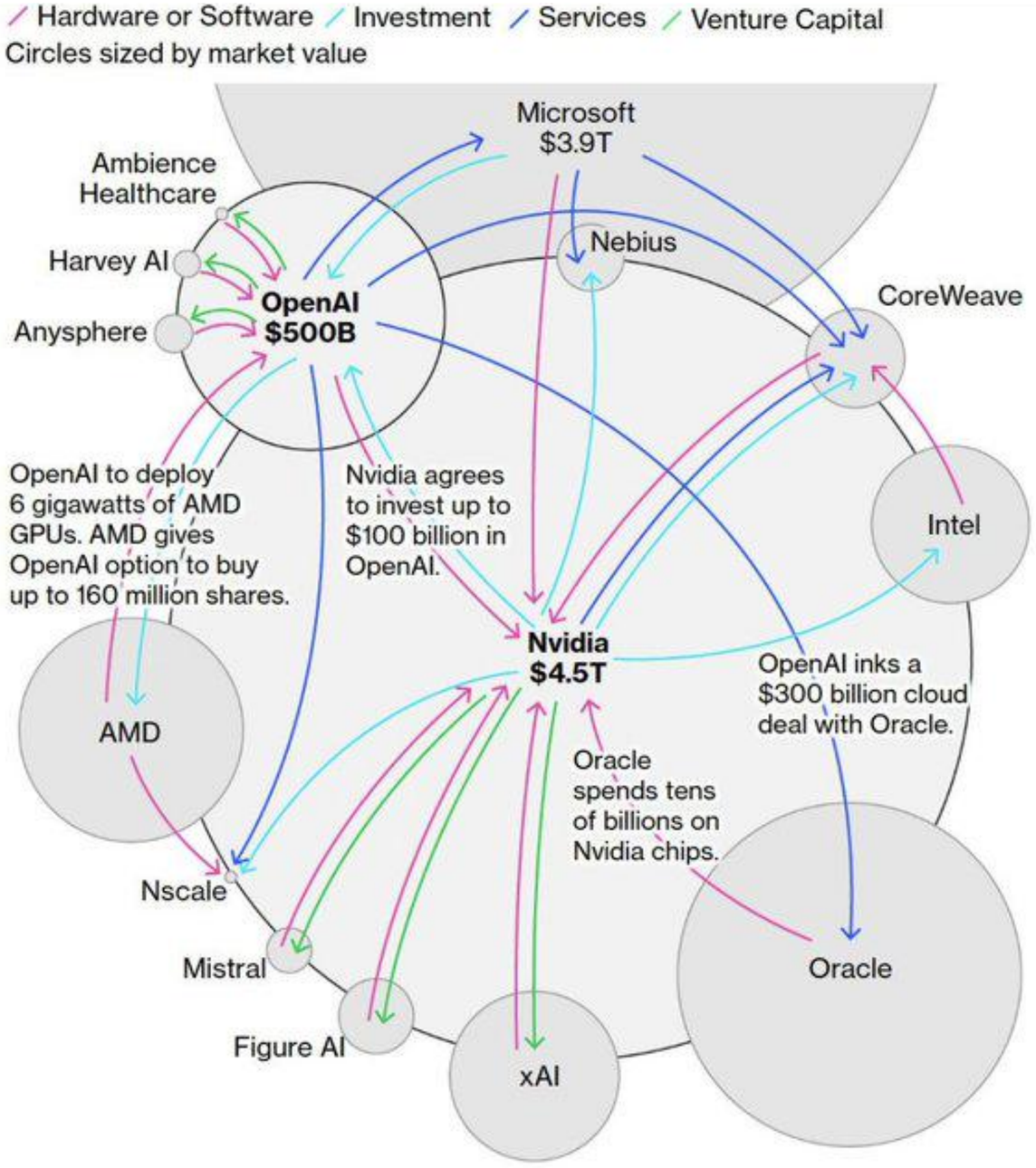
OpenAI says it plans to report stunning annual losses through 2028—and then turn wildly profitable just two years later

By **Dave Smith**
Editor, U.S. News

Add us on

November 12, 2025, 11:18 AM ET

roughly $74 billion in operating losses in 2028 alone


Hardware or Software
Investment
Services
Venture Capital
Circles sized by market value
Microsoft
$3.9T
Ambience
Healthcare
Harvey AI
Anysphere
OpenAI
$500B
Nebius
CoreWeave
OpenAI to deploy
6 gigawatts of AMD
GPUs. AMD gives
OpenAI option to buy
up to 160 million shares.
Nvidia agrees
to invest up to
$100 billion in
OpenAI.
Intel
Nvidia
$4.5T
OpenAI inks a
$300 billion cloud
deal with Oracle.
AMD
Oracle
spends tens
of billions on
Nvidia chips.
Nscale
Mistral
Figure AI
xAI
Oracle
Source: Bloomberg News reporting

Bank of England warns of AI bubble risk

Google boss says trillion-dollar AI investment boom has 'elements of irrationality'

Alphabet boss Sundar Pichai has warned of some "irrationality" in the current AI boom

If the AI bubble bursts, what will it mean for research?

The rise in artificial-intelligence technologies is unprecedented, but some predict a stock-market crash that could have knock-on effects for funding and jobs.

Artificial intelligence (AI)

Analysis

The question isn't whether the AI bubble will burst – but what the fallout will be

Eduardo Porter

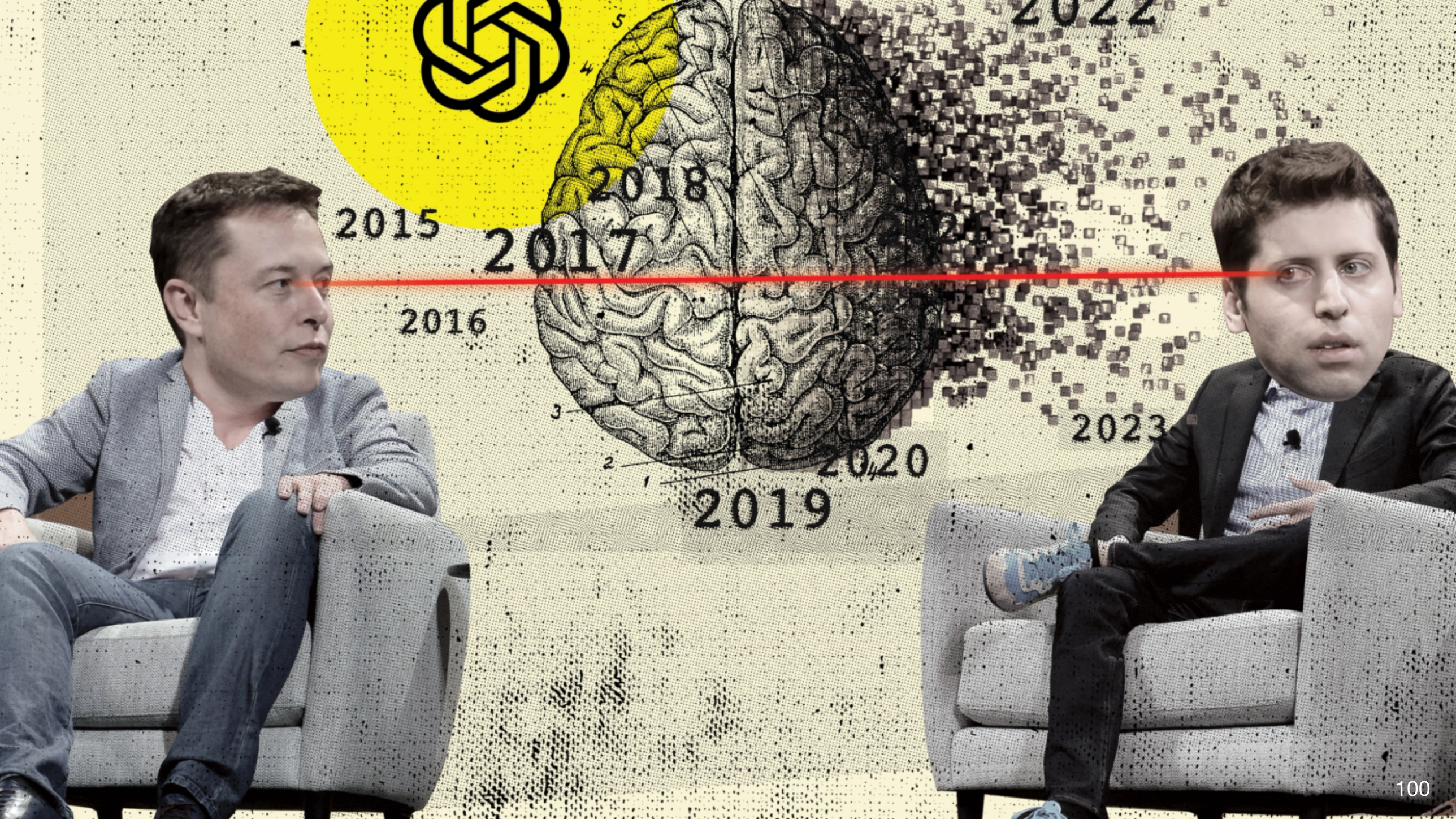
Will the bubble ravage the economy when it bursts? What will it leave of value once it pops?

Wall Street Is Shaking Off Fears of an A.I. Bubble. For Now.

The valuations of some artificial intelligence companies are approaching those of the dot-com boom. But investors worry that pulling money from today's market risks future gains.

Turning OpenAI Into a Real Business Is Tearing It Apart

Executives and researchers have left this year amid disputes over the company's values and fights among its leaders


2015
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2017
2018
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2020
2022
2023

Elon Musk's Lawsuit Could Cost OpenAI More Than $1 Billion

By Rocket Drew

Elon Musk's xAI accuses OpenAI of stealing trade secrets in new lawsuit

Suit alleges OpenAI has a 'troubling pattern' of hiring former xAI workers to access secrets about the Grok chatbot

Musk firms sue Apple and OpenAI, alleging they hurt competition

The truth about Elon Musk and OpenAI

Elon is trying everything he can to slow down OpenAI for his personal benefit. Here, we present the facts.

OpenAI sues Elon Musk claiming 'bad-faith tactics'

OpenAI Newsroom
@OpenAINewsroom · **Follow**

Today we responded to Elon's latest harassment tactic dressed up as a lawsuit. OpenAI doesn't need or want anyone's trade secrets. We will protect our employees and won't be intimidated by his attempts to bully them. openai.com/elon-musk/

ROBOTS
SIRIUS CYBERNETICS CORP.
(MARKETING DIVISION)

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A BUNCH OF MINDLESS JERKS WHO'LL
BE THE FIRST AGAINST THE WALL
WHEN THE REVOLUTION COMES.

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Questions?